

Senior High School



Lesson Exemplar in General Mathematics

Quarter 2

Unit

5

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Lesson Exemplar for General Mathematics
Quarter 2: Unit 5

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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR					
Learning Area	GENERAL MATHEMATICS			Grade Level	11
Semester	FIRST	Quarter	SECOND	Unit	5

I. OBJECTIVES	
Content Standard	The learners demonstrate knowledge and understanding functions and its graphs.
Performance Standard	The learners are able to illustrate graphically the functions involving linear and quadratic and be able to solve real-life problems such as but not limited to fare rates, purchasing, and income tax.
Learning Competencies	The learners ... 1. classify function based on their graphs and/or equations to determine if it is cubic, square root and cube root. 2. sketch the graph of the cubic, square root and cube root functions with or without technology. 3. apply the functions to model and solve real-life problems. 4. classify function based on their graphs and/or equations to determine if it is linear or quadratic. 5. sketch the graph of the linear and quadratic functions with or without technology. 6. apply the linear and functions to model and solve real-life problems. 7. illustrate a piecewise function in practical contexts (i.e. fare rates, purchasing, and income tax). 8. solve practical problems involving piecewise functions
II. REFERENCES and MATERIALS	Textbooks Demana, Waits, Clemens . Teacher's Edition <i>Precalculus Mathematics: A Graphing Approach</i>. Addison-Wesley Publishing Company, 2000. Alferez, Merle. <i>MSA Elementary Algebra</i>, MSA Academic Advancement Institute, 2007.

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Semyanko N. *Graphing Square Root & Cube Root Functions*, Algebra I, 2023

Khurma M. *Cubic and Cube Root Function*, Cue Learn Private Limited, 2011

CHED-PNU General Math Teaching Guide

Orines, F. (2016). Next Century Mathematics. General Mathematics for Senior High School. Phoenix Publishing House.

Santos, D & Biason, M. G. (2016). Math Activated: Engage Yourself and Our World. General Mathematics. Senior High School. SALESIANA BOOKS.

Stewart, J. (2016). *Precalculus: Mathematics for calculus* (7th ed.). Cengage Learning.

Video lessons

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<https://www.youtube.com/watch?v=jv8O5iKBlls>

<https://www.turito.com/learn/math/piecewise-function-grade-10>

<https://www.youtube.com/watch?v=XVdVFQGf3aY>

<https://www.youtube.com/watch?v=OYOXMyFKotc>

<https://www.youtube.com/watch?v=1WFhDqOpjpU>

Application Software

Desmos. *Desmos Scientific Calculator*. Desmos, <https://www.desmos.com/scientific>. Accessed 9 May 2025.

Geogebra. *GeoGebra Graphing Calculator*. Geogebra, <https://www.geogebra.org/graphing>. Accessed 2 October 2024.

Materials

- Desmos Calculator
- Visual aids/PowerPoint presentation
- Worksheets
- Task cards

Lesson 5.1		
III. CONTENT	Linear and Quadratic Functions	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: - recall the definition of function, domain and range; - identify and classify the function types from their equations as linear and quadratic based on their forms. - identify and classify the function types from the given graphs by analyzing their shapes and key features to determine if they represent a linear or quadratic function. - sketch the graph of the following functions: linear and quadratic. - apply the linear and quadratic functions to model and solve real-life problems.	
IV. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1.1: Pretest on Functions Instructions: Choose the letter of the correct answer. - Which of the following best defines a function? - A relation where each input has exactly one output - A relation where inputs can have multiple outputs - A set of ordered pairs - A graph that forms a straight line Answer: A - What is the slope of the linear function $f(x) = 5x - 2$? -2 5 2 -5 Answer: B	Activity A.1.1: Pretest on Functions To activate learners' prior knowledge on the topic, begin with a pretest using a multiple-choice questions. To facilitate this activity, ask the learners to write their answers on a clean sheet of paper. Review the pretest by explaining the correct answers and addressing any misconceptions

3. Which of the following represents a quadratic function?

A. $y = 3x + 5$

B. $y = 3x^2 - 2x + 1$

C. $y = \frac{1}{x}$

D. $y = x^3$

Answer: B

4. What is the shape of the graph of a quadratic function?

A. Line

B. Circle

C. Parabola

D. Triangle

Answer: C

5. What is the y-intercept of the line $y = -2x + 5$?

A. -2

B. 5

C. -5

D. 2

Answer: B

Ready for a challenge? Let's jump into a fun activity to test what you know! Think of it as a brain warm-up before we dive deeper into the lesson.

Activity A.1.2: Function Charades

Setup:

Activity A.1.2: Function Charades

Function Charades: Game Mechanics

Setup:

1. **Write a list of terms/concepts related to functions on index cards or small pieces of paper. Examples include:**

	- Write a list of terms/concepts related to functions on index cards or small pieces of paper. Examples include: - Function - Domain - Range - Graph - Linear function - Quadratic function - Slope - Vertex - Divide the class into two or more teams. - Assign designated area at the front of the room for the acting. Instructions: - One player from a team draws a card and acts out the term without speaking. They can use gestures, body movements, or props. - Their teammates have 60 seconds to guess the term. - If the team guesses correctly within the time limit, they earn a point. - The charade shall to the next team, after 60 seconds. Examples of Charades Actions: Function: Act as a "function machine": pretend to take an input (e.g., a ball), process it (e.g., shake it in your hands), and produce an output (e.g., another object like a pen).	Prepare the index cards or small pieces of paper in advance. Write one term or concept on each card. Ensure the terms are clearly written and large enough to be read easily. Examples: - Function - Domain - Range - Graph - Linear function - Quadratic function - Slope - Vertex 2. Divide the class into two or more teams. Organize the learners into teams of equal size. This will ensure fair play and encourage teamwork. 3. Provide a designated area at the front of the room for the acting. Clear a space at the front of the room where players can act out the terms. Make sure there is enough room for movement and that all learners can see the acting area. Instructions: One player from a team draws a card and acts out the term without speaking. They can use gestures, body movements, or props. Explain the rules clearly to the learners. Emphasize that the player acting out the term cannot speak or use any written
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	2. Domain: Point to a row of imaginary "input numbers" and pretend to select them to feed into a function machine. 3. Range: Mimic receiving "output numbers" after processing inputs in a machine. Gesture to indicate the results. 4. Graph: Pretend to draw a graph on an imaginary whiteboard or mimic plotting points on a grid. 5. Linear Function: Stretch your arms out in a straight line to show the constant slope of a linear graph. 6. Quadratic Function: Use your arms to form a "U" shape, representing a parabola. 7. Slope: Pretend to climb or slide down an imaginary hill, emphasizing the steepness of the incline. 8. Vertex: Point to the lowest or highest point of an imaginary parabola you draw in the air.	words. They should rely on gestures, body movements, and props if available. 2. Their teammates have 30–60 seconds to guess the term. Set a timer for 30–60 seconds for each round. Adjust the time limit based on the difficulty of the terms and the learners' familiarity with the concepts. 3. If the team guesses correctly within the time limit, they earn 5 points. Keep track of the points for each team. You can use a whiteboard or a scoreboard to display the scores. 4. Rotate to the next team. Ensure that each team gets an equal number of turns. Rotate the teams in a consistent order to maintain fairness Note: If the time permits, you can conduct the pretest activity and function charade activity.
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	A.2. Establishing the Purpose of the Lesson Activity A.2: Quick Plot Challenge Instructions Get Ready: Bring out graphing paper, ruler, and pencil. Graph These Functions: $y = 2x - 1$ $y = x^2 + 2$ - Choose 2 to 4 points values for x (- Find the matching y values for each function. - Plot the points on your graph. Label Everything: - Write the name of each function as the title. - Label the x-axis and y-axis. - Mark at least 3 important points (like where the graph crosses the axes or the highest/lowest point). Check Your Work: - Make sure your graph is neat and correct. - Compare with a classmate or ask your teacher for feedback. Make a short discussion on graphing linear functions and quadratic functions quickly.	Activity A.2: Quick Plot Challenge To facilitate this activity, follow the instructions. Instructions: - The teacher will pair each learner. Note that pairing depends on how the teacher does it. It could be by random selection, seating arrangement, etc. - Each item in the activity shall be flashed on the screen and let all pairs think of the activity. - The teacher will tell the class about the activity called the Quick Plot Challenge using the “Think-Pair-Share”. - The teacher shall discuss the purpose of the activity by saying “<i>To classify functions by inspection correctly, you need to understand the key characteristics of functions such as their properties, graphs, behavior. This technique will help you recognize functions easily and accurately.</i>” - To facilitate the activity, the teacher shall: ✚ Tell the learners first to identify the function individually, ✚ Ask them to justify their answers as a pair. ✚ Share responses as a class to clarify misconceptions. ✚ Discuss the shortcut ways of graphing linear functions and quadratic functions. The teacher shall summarize all the responses given and provide supplemental discussions <i>Note:</i> For low-performing learners: Provide step-by-step guidance, ensuring they understand how to use the table values for plotting.
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Shortcut for Graphing Linear Functions

1. Start with the y-intercept:

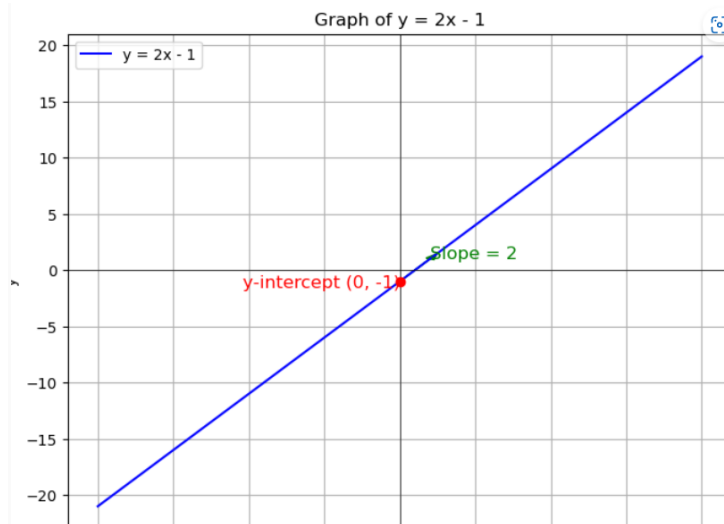
- The equation is in slope-intercept form: $y = mx + b$
- Here, $b = -1$, so plot the point **(0, -1)** on the y-axis.

2. Use the slope:

- The slope $m = 2$ means **rise 2, run 1**.
- From (0, -1), move **up 2 units** and **right 1 unit** to plot the next point: **(1, 1)**.

3. Draw the line through the points.

Here is the graph of the function $y = 2x + 1$



Shortcut for Graphing $y = x^2 + 2$

1. Identify the vertex:

- The function is in the form $y = x^2 + c$, so the vertex is at **(0, 2)**.

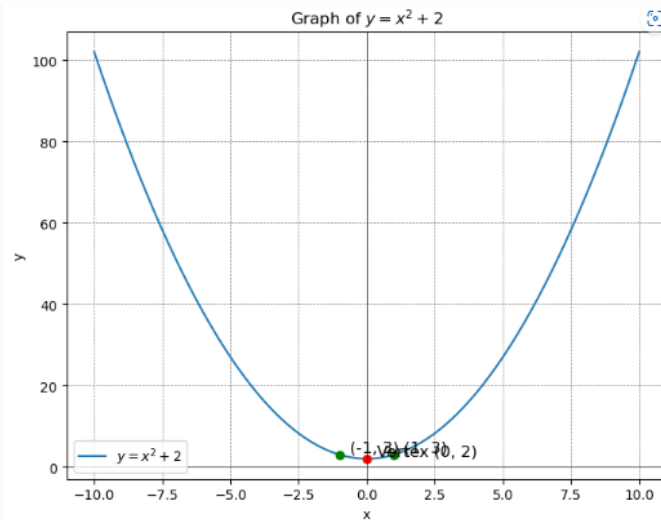
2. Plot symmetric points:

For high-performing learners: Ask them to predict the graph's slope and intercept before plotting or challenge them to derive the equation from the graph.

Employ technology integration: For tech-savvy learners, let them use online tools (e.g., Desmos) to check their work.

- Choose values of x around the vertex (e.g., $x = -1$ and $x = 1$):
 - $y = (-1)^2 + 2 = 3$
 - $y = (1)^2 + 2 = 3$
 - Plot the points **$(-1, 3)$** and **$(1, 3)$** .
3. **Draw the parabola:**
- Sketch a smooth curve through the vertex and symmetric points.

Here is the graph of the function $y = x^2 + 2$



B. Instituting New Knowledge

B.1. Presenting Examples

Activity B.1: Classifying the Functions as Linear or Quadratic Through Graphs

Instructions:

Classify the Function – Linear, Quadratic, or Neither

1. Divide the class into five groups.

Each group will receive a set of mini whiteboard answer cards.

Activity B.1: Classifying the Functions as Linear or Quadratic Through Graphs

Materials Needed

- Graph cards or slides
- Mini whiteboards or answer cards

Instructions:

2. **Watch the screen carefully.**
Graphs will be shown one at a time through a PowerPoint presentation.
3. **You have 20 seconds** to analyze each graph and decide whether it represents a **linear**, **quadratic**, or **neither** function.
4. **After 20 seconds**, each group must raise their answer card.
5. **Scoring:**

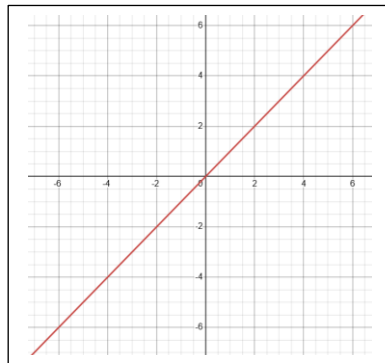
☑ Correct answer = **2 points**

✗ Incorrect answer = **0 points**

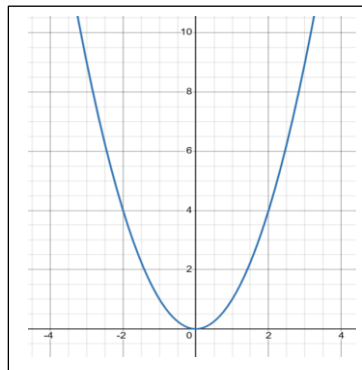
6. **The group with the highest total score at the end of the game wins!**

Here are the sample graphs.

Graph 1



Graph 2



1. **Here are the samples of the graphs that will help learners determine whether it is linear or quadratic functions:**

Provide learners with visual examples of both linear and quadratic functions. Ensure the graphs are clearly labeled and easy to interpret.

Linear Function (Line):

Explain that a linear function is represented by a straight line on the graph. The general form of a linear function is $y = mx + b$, where m is the slope and b are the y-intercept.

Quadratic Function (Parabola):

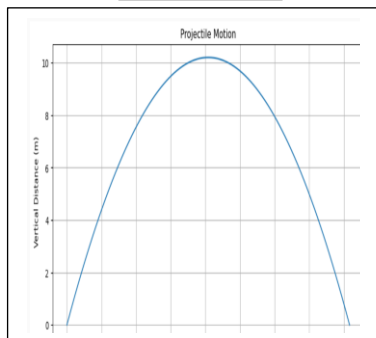
Explain that a quadratic function is represented by a parabola on the graph. The general form of a quadratic function is $y = ax^2 + bx + c$, where a , b , and c are constants, and $a \neq 0$.

2. **Ask learners to classify the given graphs as either linear or quadratic.**

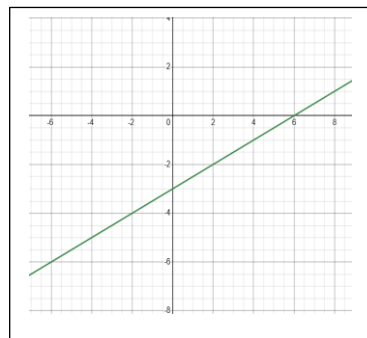
Provide a set of graphs for the learners to analyze. Ensure there is a mix of both linear and quadratic functions. Ask learners to identify and classify each graph based on its shape.

3. **Discuss the characteristics of linear and quadratic functions.**

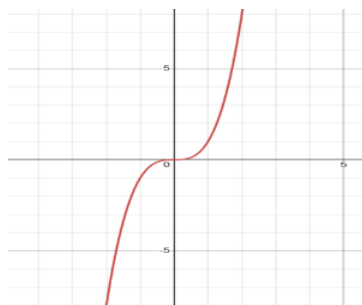
Graph 3



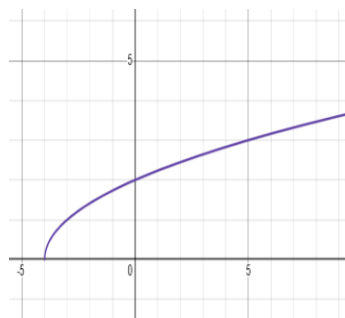
Graph 4



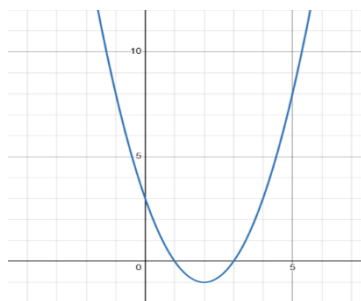
Graph 5



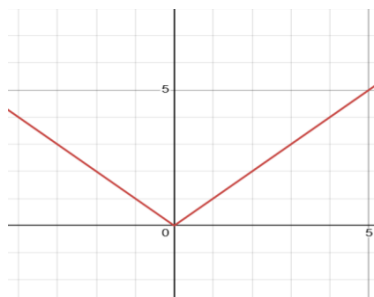
Graph 6



Graph 7



Graph 8



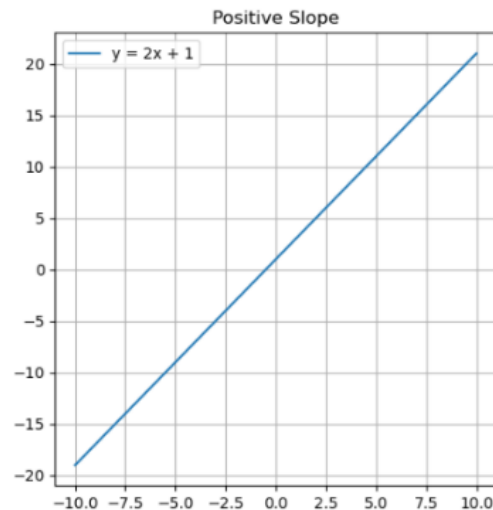
Facilitate a discussion on the key features of linear and quadratic functions. Highlight the differences in their graphs, equations, and real-life applications.

	Answer: By looking at the graphs, the correct answer is linear functions are graphs 1 and 4 and quadratic functions are graphs 2 and 3	
	B.2. Discussing the Concept Activity B.2 Lesson Discussion: Linear Functions and Quadratic Functions Instructions: Study the information and examples and prepare for class discussion. Key Characteristics of Linear and Quadratic Functions Linear Functions 1. Equation Form: The general form of a linear function is $y = mx + b$, where: ▪ m is the slope. ▪ b is the y – intercept. 2. Graph Shape: The graph of a linear function is a straight line. 3. Slope: The slope (m) indicates the steepness and direction of the line. ▪ Positive slope: Line rises from left to right.	Activity B.2 Lesson Discussion: Linear Functions Instructions: In order for learners to gain a better understanding of financial mathematics, key terms need to be clearly defined and thoroughly explained. To facilitate the discussion, you may apply any of the following strategies. Strategy 1: Fishbowl Discussion 1. Prepare strips of paper with different terms or equations written on them ($y = mx + b$, <i>slope</i>, <i>straight line</i>, <i>vertex parabola</i> and other concepts on linear and quadratic functions) and place them inside a box. 2. Randomly pick a strip of paper from the box. 3. Using the same groupings, assign one member from each group to pick a term and provide their own explanation or definition of the term while the rest of the group listens attentively. 4. Repeat the process for each group, allowing different members to explain. 5. Assist learners who may struggle to define or explain their terms by providing guiding questions or hints. 6. Facilitate an interactive discussion, encouraging groups to ask questions, add examples, or clarify meanings to deepen their understanding. Strategy 2: Cabbage Paper Passing 1. Write different terms on 5 small pieces of paper.

- Negative slope: Line falls from left to right.
- Zero slope: Line is horizontal.
- Undefined slope: Line is vertical.

1. Example: Linear function with positive slope:
 $y = 2x + 1$

The line rises from left to right.
 Slope = 2, Y-intercept = 1.

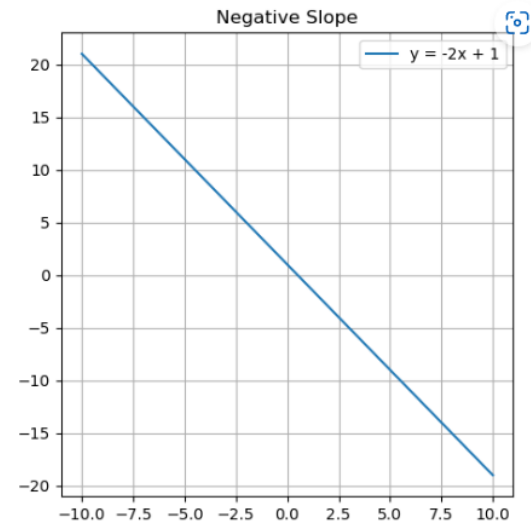


Example: Linear function with negative slope:

$$y = -2x + 1$$

The line falls from left to right.
 Slope = -2, Y-intercept = 1.

2. Crumple them together into a ball, like a cabbage with layers.
3. Give one cabbage to each group.
4. Play music while learners pass the cabbage around.
5. When the music stops, the learner holding it opens one layer and explains the word.
6. Keep going until all layers are opened and explained.
7. The teacher helps and encourages learners during the activity.

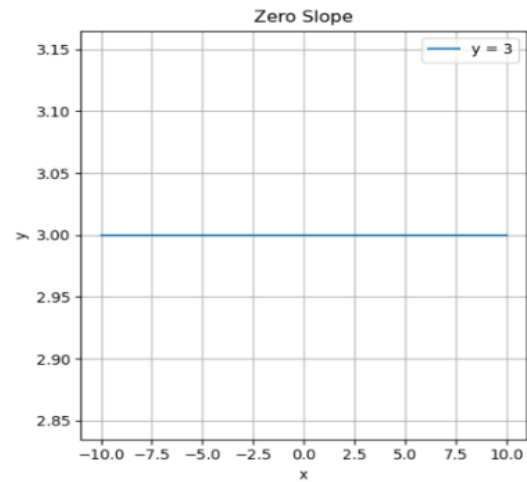


Example: Linear function with zero slope:

$$y = 3$$

The graph is a horizontal line.

The Slope = 0 and y-intercept = 3.

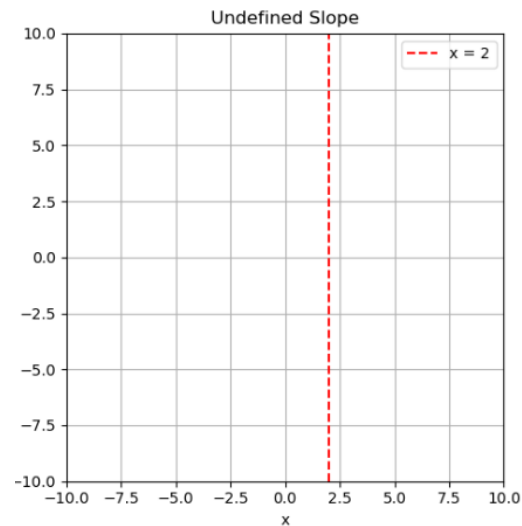


Example: Linear function with undefined slope:

$$x = 2$$

The graph is a vertical line.

Slope is undefined, passes through $x = 2$.



4. **Y-Intercept:**

The **y-intercept** (b) is the point where the line crosses the y-axis.

5. **Rate of Change:**

Linear functions have a **constant rate of change**.

The rate of change is the slope of the linear function.

Example1: In the given function $y = 3x + 2$, find the slope and the y-intercept. Then sketch the graph.

Solutions:

Step 1: Identify the slope and y-intercept

Slope (m): The coefficient of (x) is **3**.

Y-intercept (b): The constant term is **2**, meaning the graph crosses the y-axis at $((0, 2))$.

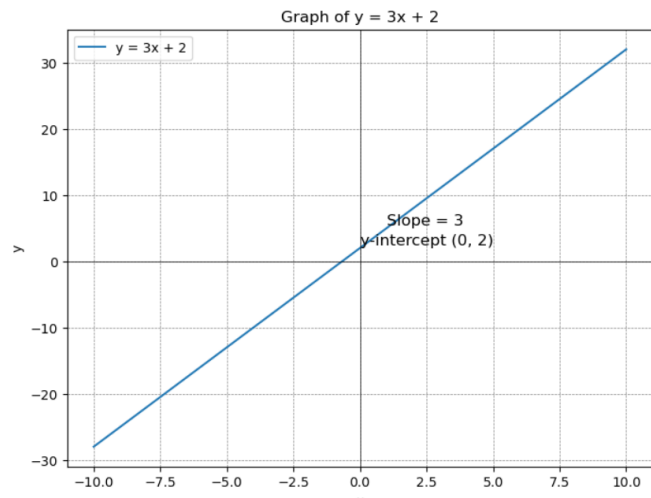
Step 2: Sketch the graph

Start at the **y-intercept** $(0, 2)$.

Use the **slope** to find another point:

Since the slope is 3, go **up 3 units** and **right 1 unit** from the y-intercept to get the point $(1, 5)$.

Draw a straight line through these points.



Example 2: Graph the function $y = 2x + 1$.

Answer:

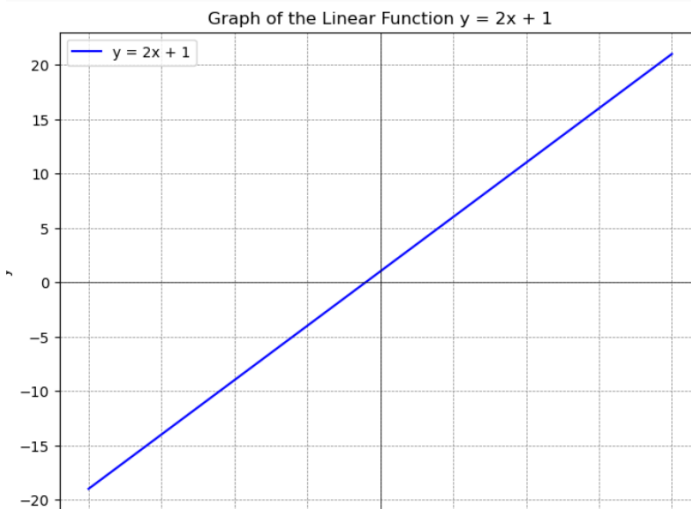
Note:

Make sure to clearly explain the solutions to each example. Use tools like GeoGebra, Desmos, or other graphing apps to help visualize and deepen learners' understanding.

For Example 1: Determine the Slope and the Y-Intercept of the Function $y = 2x + 1$.

Instruction:

1. **Determine the slope and the y-intercept of the function $y = 2x + 1$.**



Activity: YOUR TURN

Instructions:

Graph the following functions using the shortcut ways in graphing and answer the questions that follow.

a. $y = -2x + 2$

b. $y = 3x - 4$

c. $2x + y + 4 = 0$,

Guide Questions:

1. What do you notice on the graphs of the three functions?
2. How do you determine the points on the third function?

To find the slope and the y-intercept, compare the given function to the general form of a linear equation $y = mx + b$ where:

- m is the slope.
- b is the y-intercept.
- **Slope (m):**

In the function $y = 2x + 1$, the coefficient of x is 2. Therefore, the slope (m) is 2.

- **Y-Intercept (b):**

In the function $y = 2x + 1$, the constant term is 1. Therefore, the y-intercept (b) is 1.

The equation in the form $y = mx + b$, m is the value of the slope and b is the y-intercept. This is how to find the slope and y-intercept in the Example 1.

Example 2: Graph the Function $y = 2x + 1$.

Instructions:

1. **Graph the function $y = 2x + 1$.**

To graph the function, follow these steps:

Step 1: Identify the slope and y-intercept.

From the equation

Discuss the Properties of Quadratic Functions

1. Equation Form:

The general form of a quadratic function is $y = ax^2 + bx + c$, where:

(a), (b), and (c) are constants and a is not equal to zero.

(a) determines the **direction** and **width** of the parabola.

2. Graph Shape:

The graph of a quadratic function is a **parabola**.

3. Vertex:

The **vertex** is the highest or lowest point on the graph, depending on the direction of the parabola.

- If ($a > 0$), the parabola opens upwards, and the vertex is the **minimum** point.
- If ($a < 0$), the parabola opens downwards, and the vertex is the **maximum** point.

4. Axis of Symmetry:

- The **axis of symmetry** is a vertical line that passes through the vertex, given by

$$y = 2x + 1:$$

- The slope (m) is 2.
- The y-intercept (b) is 1.

Step 2: Plot the y-intercept on the graph.

The y-intercept is the point where the line crosses the y-axis. Plot the point (0, 1) on the graph.

Step 3: Use the slope to find another point.

The slope of 2 means that for every 1 unit increase in x, y increases by 2 units. Starting from the y-intercept (0, 1), move 1 unit to the right (to $x=1$) and 2 units up (to $y=3$). Plot the point (1, 3).

Step 4: Draw the line.

Draw a straight line through the points (0, 1) and (1, 3). Extend the line in both directions.

Note:

For Activity: Your Turn, let the learner's device their own way to graph the linear functions. They can use the shortcut way or graphing applications.

Ask the learners on the guide questions,

$$x = -\frac{b}{2a}$$

5. **Y-Intercept:**

- The **y-intercept** is the point where the parabola crosses the y-axis, found by evaluating (y) when (x = 0).

6. **Direction and Width:**

- The coefficient (a) affects the **direction** and **width** of the parabola.
 - Larger absolute value of (a): Narrower parabola.
 - Smaller absolute value of (a): Wider parabola.

7. **Roots/Zeros:**

- The **roots** or **zeros** of the quadratic function are the points where the graph intersects the x-axis, found by solving $ax^2 + bx + c = 0$.

Example 3: Work in group to construct the graph of $y = x^2 - 4x + 3$ in large poster.

Instructions:

1. Divide the class into small groups of 3-5 learners.
2. In a Large Poster, graph the function $y = x^2 - 4x + 3$
3. You are free to use any techniques in graphing.
4. Each group presents their graph, explaining the process and features of the parabola.

Answer: Sample graph of $y = x^2 - 4x + 3$.

Example 3: Work in Group to Construct the Graph of $y = x^2 - 4x + 3$ on a Large Poster

Instructions:

1. **Group Formation:**

Divide the class into small groups of 3-5 learners. Ensure each group has a mix of abilities to promote collaborative learning.

2. **Step 1: Table of Values**

Ask each group to create a table of values for x ranging from -1 to 5. This will help them plot the points accurately.

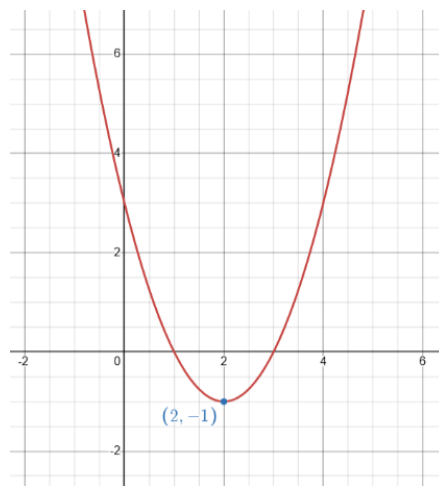
Calculate y values using the equation $y = x^2 - 4x + 3$.

Step 2: Plotting Points

On the large poster, draw a coordinate plane. Ensure the axes are labeled and scaled appropriately.

- **Plot the points from the table of values accurately.**

3. **Step 3: Drawing the Parabola**



For better understanding elaborate the easy way to graph the quadratic functions using the vertex form:

Vertex Form of a Quadratic Function

$$y = a(x - h)^2 + k$$

Where:

- (h, k) is the **vertex**
- (a) determines the **direction** and **width** of the parabola
- **Shortcut Steps to Graph**

Step 1: Identify the Vertex

- From the equation, extract (h) and (k)
- Example: $(y = (x - 3)^2 + 2) \rightarrow$ Vertex is $(3, 2)$

Connect the points smoothly with a curve to form the parabola. Emphasize the importance of accuracy in connecting the points.

Highlight key features such as:

- **Vertex:** $(x = 2, y = -1)$
- **Axis of symmetry:** $x = 2$
- **X-intercepts:** $x = 1$ and $x = 3$
- **Y-intercept:** $y = 3$

4. Step 4: Labeling and Analyzing

Label the graph with:

- **Equation:** $y = x^2 - 4x + 3$
- **Key points:** (vertex, intercepts)
- **Axis of symmetry**

5. Step 5: Group Presentation

Each group presents their graph, explaining the process and features of the parabola. Encourage learners to discuss how they calculated the points and identified the key features.

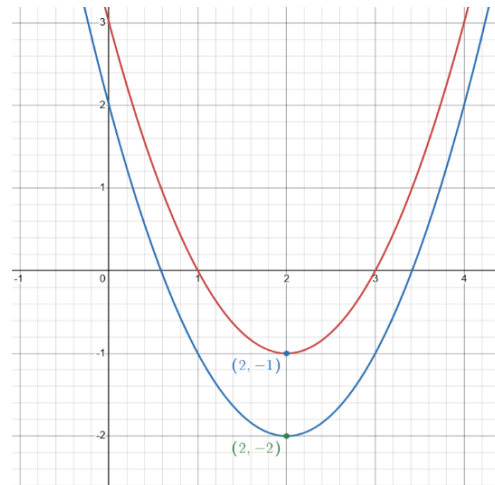
	Step 2: Plot the Vertex - This is the turning point of the parabola. Step 3: Determine the Direction - If ($a > 0$): Parabola opens upward - If ($a < 0$): Parabola opens downward Step 4: Use “1-3-5” Rule for Standard Width - From the vertex, move: 1 unit left/right → 1 unit up/down 2 units left/right → 4 units up/down 3 units left/right → 9 units up/down - Multiply these values by (a) if ($a \neq 1$). Step 5: Draw the Axis of Symmetry - A vertical line through ($x = h$) Step 6: Sketch the Parabola - Plot symmetric points on both sides of the vertex - Connect the points with a smooth curve Note if quadratic equation in standard form $y = ax^2 + bx + c$, convert into vertex form $y = a(x - h)^2 + k$. Show the steps of converting standard form into vertex form.	Example 4: Instructions: Graphing Quadratic Functions Objective: - To graph the function $y = x^2 - 4x + 2$ - To compare it with the graph of $y = x^2 - 4x + 3$. Steps: Graphing the First Function $y = x^2 - 4x + 2$: - Plot the vertex, axis of symmetry, and a few additional points. - Identify the parabola's direction, vertex, and y-intercept. - Use graphing applications such as Desmos Calculator or GeoGebra for accuracy. - Plot the vertex, axis of symmetry, and a few additional points. - Identify the parabola's direction, vertex, and y-intercept. Graphing the Second Function $y = x^2 - 4x + 3.$ - Overlay the graph of $y = x^2 - 4x + 3$ on the same coordinate plane. - Highlight the changes in the vertex and y-intercept compared to the first function. Comparison Guide: Similarities: Both functions have the same shape (parabola) and symmetry, as they share the $x^2 - 4x$ term.
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Watch the steps in the YouTube video: [Introduction to vertex form of a quadratic](https://www.youtube.com/watch?v=_QqhuLixNEk)

https://www.youtube.com/watch?v=_QqhuLixNEk

Example 4: Graph the function $y = x^2 - 4x + 2$. Then compare the graph of $y = x^2 - 4x + 3$.

Answer:



The comparison of the graphs of $y = x^2 - 4x + 2$ and $y = x^2 - 4x + 3$:

A. Vertex:

- Both functions have the same vertex x - coordinate, which is $x = 2$.
- The y-coordinates of the vertices are different:
For $y = x^2 - 4x + 2$, the vertex is at $(2, -2)$
For $y = x^2 - 4x + 3$, the vertex is at $(2, -1)$

- b. **Differences:** Observe the shift in the graph's position, particularly the y-intercept and how it affects the vertex.

4. Exploration Questions for Learners:

- What happens to the graph when the constant term changes?
- How does the vertex shift between the two graphs?
- What remains unchanged between the two graphs, and why?

5. Interactive Use of Tools:

Encourage learners to use sliders in applications like Desmos to visualize how changes in the constant term affect the graph.

Note:

Allow the learners to use graphing applications. Let the learner browse the graphing apps like geogebra or desmos calculator. If they are not familiar with the apps, the teacher will teach the learners on how to use the graphing applications.

	B. Shape and Direction: - Both graphs are parabolas that open upwards because the coefficient a is positive in both cases. C. Y-intercept The y-intercepts are different: - For $y = x^2 - 4x + 2$, the y-intercept is (0,2) - For $y = x^2 - 4x + 3$, the y-intercept is	
	B.3. Developing Mastery Activity B.3: Practice: Group Work, Graph the following functions: $y = 2x + 1$ $y = x - 4$ $y = -\frac{1}{2}x + 3$ $y = x^2 - 4x + 3$ $y = -x^2 + 2x + 1$ $y = x^2 + x - 6$ $Y = -2x^2 - 3x + 1$	Activity B.3: Practice: Group Work, Note: Give the following instructions to the learners. Objective of the Activity: Work as a team to graph and analyze one assigned function. This activity will help you understand how different types of functions behave and how to represent them visually. Instructions: 1. Your group will be assigned one function from the list below: $y = 2x + 1$ $y = x - 4$ $y = -\frac{1}{2}x + 3$ $y = x^2 - 4x + 3$ $y = -x^2 + 2x + 1$ $y = x^2 + x - 6$ $y = x^2 + x - 6$ $Y = -2x^2 - 3x + 1$ 2. Materials Needed: - Graphing paper or printed grid sheets - Ruler - Pencil and eraser

		d. Colored pens or markers (optional) 3. Graphing Guidelines: e. Label the x- and y-axes clearly. f. Plot at least 5 points for your function. g. Use a ruler for straight lines and a smooth curve for parabolas. h. Label your graph with the equation. i. Identify and label key features: i. For linear functions: slope and y-intercept ii. For quadratic functions: vertex, axis of symmetry, and intercepts 4. Collaboration: j. Work together to analyze the function and plot it accurately. k. Make sure everyone in the group understands the process. l. Assign roles (e.g., plotter, checker, explainer, presenter). 5. Presentation: m. Be ready to present your graph to the class. n. Explain how you graphed it and describe its key features. 6. Submission: o. Submit your group's graph with the names of all members.
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C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Solve the following problems. Problem 1: Know your Household Consumption A household pays a fixed monthly connection fee of ₱300 and ₱10 per kilowatt-hour of electricity consumed. • Write a linear function for the monthly bill B based on the number of kilowatt-hours k used. • Calculate the bill if the household consumes 150 kilowatt-hours. Linear Function: Answer: $B(k) = 300 + 10k$ Total Bill : Solutions: Used the linear function equation $B(k) = 300 + 10k$, Where B is the constant and k is the number kilowatt-hours used. By substitution method, when $k = 150$, $B(150) = 300 + 10(150)$ $B(150) = 300 + 1500$ $B = 1800$ The bill of the household is 1800 pesos. Problem 2:	To facilitate the discussion, teacher may select any of the following strategies: Pedagogy: Problem-Based Learning (PBL), Strategy 1: Think-Pair-Share (Collaborative Problem Solving) Steps: 1. Think – Learners work individually to solve a linear or quadratic problem (e.g., write a function or solve an equation). 2. Pair – Learners pair up to compare their approaches and solutions. 3. Share – Pairs share their reasoning with the class; the teacher facilitates discussion and highlights key strategies. Strategy 2: 4-Step Problem Solving Model (Polya's Method) Steps: 1. Understand the Problem – Identify what is known and what needs to be found. 2. Devise a Plan – Choose a strategy (e.g., write an equation, graph, use a formula). 3. Carry Out the Plan – Solve the problem step-by-step. 4. Look Back – Check the solution and reflect on the method used. Strategy 3: Real-Life Application Projects
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	Scenario: Planning a school fundraiser. 1. Situation: You are selling tickets for a fundraising event at a fixed price of Php 100 per ticket. The fixed cost to organize the event is Php 5,000. 2. Activity: - o Ask learners to calculate the profit for selling 50, 100, 200, and 350 tickets. Solutions: Use the profit function $P(x) = 100x - 5000$, where x is the number of tickets sold. Profit for selling 50 tickets: $P(50) = 100(50) - 5000 = 0$ Profit for selling 100 tickets: $P(100) = 100(100) - 5000 = 5000$ Profit for selling 200 tickets: $P(200) = 100(200) - 5000 = 15000$ Profit for selling 350 tickets: $P(350) = 100(350) - 5000 = 30000$ Problem 3 : . Projectile Motion Scenario: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters. Guide Questions:	Steps: 1. Present a Scenario – E.g., electricity billing, fundraising, or projectile motion. 2. Model the Situation – Guide learners to write a linear or quadratic function. 3. Analyze and Solve – Use the function to answer questions (e.g., cost, profit, height). 4. Present Findings – Learners explain their process and results through a report or presentation. Problem 3: Projectile Motion Scenario: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters. Guide Questions: 1. At what time does the ball reach its maximum height? The maximum height is reached at t
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1. At what time does the ball reach its maximum height?

Answer: The time of maximum height is at $t = -\frac{b}{2a}$

$$t = -\frac{20}{2(-4.9)} = t = \frac{20}{9.8} \approx 2.04 \text{ seconds}$$

2. What is the maximum height of the ball?

Answer : Substitute $t = 2.04$ into $h(t)$

$$h(2.04) = -4.9(2.04)^2 + 20(2.04) + 1.5 \approx 21.1 \text{ meters}$$

3. When does the ball hit the ground?

Answer: Solve $h(t) = 0$

$$-4.9t^2 + 20t + 1.5 = 0 \text{ (use the quadratic formula)}$$

$$t \approx 0.07 \text{ or } t \approx 4.15 \text{ seconds}$$

The ball hits the ground at $t \approx 4.15 \text{ seconds}$

4. Graph the function and explain the shape of the parabola in terms of the ball's motion.

Answer:

The time of maximum height is at $t = -\frac{b}{2a}$

$$t = -\frac{20}{2(-4.9)} = t = \frac{20}{9.8} \approx 2.04 \text{ seconds}$$

Substitute $t = 2.04$ seconds into the height function $h(t)$ to find the maximum height.

Calculation: $h(2.04) = -4.9(2.04)^2 + 20(2.04) + 1.5 = 21.91$ meters.

2. **When does the ball hit the ground?**

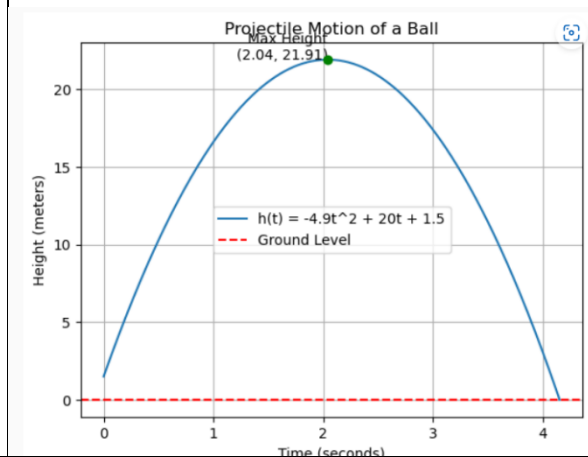
The ball hits the ground when $h(t)=0$. Solve the quadratic equation $-4.9t^2+20t+1.5=0$ to find the time t .

Calculation: The ball hits the ground at $t = 4.16$ seconds.

3. **Graph the function and explain the shape of the parabola in terms of the ball's motion.**

Plot the function $h(t) = -4.9t^2 + 20t + 1.5$ on a graph. The graph is a parabola that opens downwards, representing the ball's motion. The vertex represents the maximum height, and the x-intercepts represent the times when the ball is at ground level.

Here's the graph the of the function:



Problem 4: A small halo-halo stand in **Burnham Park, Baguio City** sells each cup for **₱40** and serves **120 customers per day**. The owner is considering increasing the price. Based on customer feedback, for **every ₱2 increase**, they expect to lose **6 customers**.

Create model of the **daily profit** as a quadratic function of the number of ₱2 price increases.

Step 1: Define Variables

Let:

- x = number of ₱2 price increases
- New price per cup = $40 + 2x$
- Number of customers = $120 - 6x$
- Profit = (price per cup) $\times$ (number of customers)

Step 2: Write the Profit Function

$$P(x) = (40 + 2x)(120 - 6x)$$

Expand:

$$\begin{aligned} P(x) &= 4800 - 240x + 240x - 12x^2 \\ P(x) &= -12x^2 + 4800 \end{aligned}$$

Step 3: Analyze the Function

This is a **downward-opening parabola**, and the maximum profit occurs at the **vertex**.

Since the linear term cancels out, the vertex is at:

$$x = 0$$

So, the **maximum profit** is:

$$P(0) = 4800$$

Conclusion:

- The **maximum daily profit** is **₱4,800**
- This occurs when the price remains at **₱40**

- Increasing the price reduces the number of customers enough to lower profit

Problem 5 : A farmer in **Nueva Ecija** wants to build a rectangular vegetable garden along the side of a **barn**, using **30 meters of fencing**. Since one side is against the barn, she only needs to fence **three sides** (two widths and one length).

What dimensions will **maximize the area** of the garden?

Step 1: Define Variables

Let:

- x = length of each **width** (the two shorter sides)
- Then the **length** (along the barn) is $30-2x$

Step 2: Write the Area Function

$$\begin{aligned} \text{Area} &= \text{length} \times \text{width} = (30 - 2x)(x) \\ A(x) &= -2x^2 + 30x \end{aligned}$$

Step 3: Find the Maximum Area

This is a quadratic function opening downward, so the **maximum** occurs at the **vertex**:

$$x = \frac{-b}{2a} = \frac{-30}{2(-2)} = 7.5$$

Substitute back to find the length:

$$\text{Length} = 30 - 2(7.5) = 15$$

Maximum Area = $15 \times 7.5 = 112.5 \text{ m}^2$ Final Answer:

- **Width** = 7.5 meters
- **Length** = 15 meters
- **Maximum Area** = 112.5 square meters

C.2. Making Generalization

Activity C.2: Compare and Conclude

Objective:
Learners will summarize the differences and similarities between linear and quadratic functions through guided questions, enabling them to generalize the key concepts.

Part 1: Complete the Table

Provide learners with the following table to fill in based on their understanding of linear and quadratic functions:

Feature	Linear Function	Quadratic Function
Graph Shape		
Equation		
Key Features		
Rate of Change		
Real-Life Examples		

Part 2: Guided Questions for Generalization

After completing the table, ask the learners the following questions to guide their generalization:

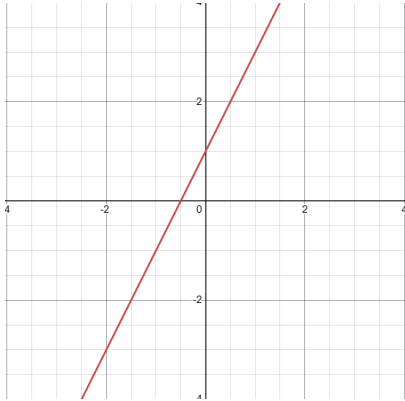
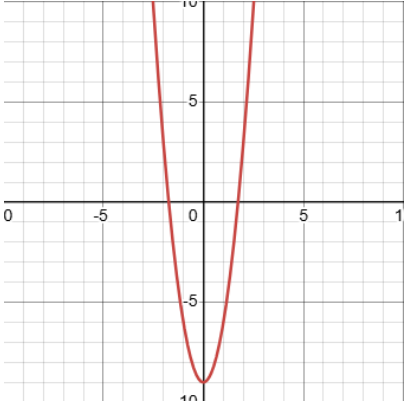
Activity C.2: Compare and Conclude

For the generalization part, let the learners do the Activity: Compare and Conclude.

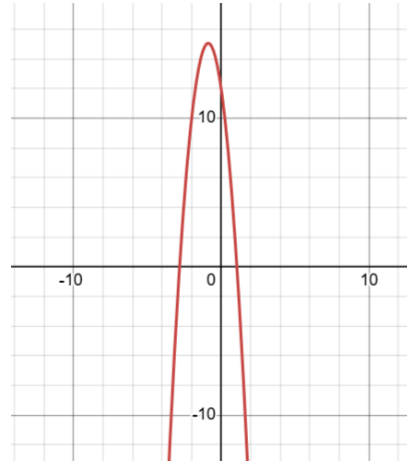
	Graph Shape: - How does the graph of a linear function differ from that of a quadratic function? - What does the shape of each graph tell you about the relationship between x and y? Equation Structure: - Compare the equations of linear and quadratic functions. What do you notice about the highest power of x? - How does the highest power of x affect the graph's shape? Key Features: - What are the key features of linear graphs? Of quadratic graphs? - Why is the slope important in a linear graph? Why is the vertex important in a quadratic graph? Rate of Change: - Describe the rate of change in a linear function. How is it different for a quadratic function? Real-Life Applications: - In what types of situations would you use a linear function? A quadratic function?	
	C.3. Evaluating Learning Activity C.3: Checking for Mastery Instructions: Analyze and solve the following: 1. Problem: A tricycle driver charges a base fare of ₱10 plus ₱5 per kilometer traveled. Question: Write a linear function that models the total fare F for traveling x kilometers. How much would it cost to travel 8 kilometers?	Activity C.3: Checking for Mastery This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

	2. Problem: A local festival in Cebu features a fireworks display. The height h of a firework in meters after t seconds is given by the function $h(t) = -4.9t^2 + 29.4t + 1.$ Question: How long does it take for the firework to reach its maximum height? What is the maximum height? Problem: Revenue Optimization for a Local Concert A learner organization in Cebu City is organizing a benefit concert. Tickets are priced at ₱300, and they expect 200 attendees at that price. They estimate that for every ₱20 increase in ticket price, 10 fewer people will attend. 1. Write a quadratic function that models the total revenue based on the number of ₱20 price increases. 2. Determine the ticket price that will maximize revenue. 3. Find the maximum possible revenue. 4. Interpret the result in the context of the problem.	
	C.4. Additional Activities Activity C.4: Reinforcement and Enrichment Remediation Graphing Linear and Quadratic Functions Problem 1: Graph the linear function $y = 3x - 2$. • Identify the slope and y-intercept. • Plot the y-intercept on the graph. • Use the slope to find another point on the line.	Activity C.4: Reinforcement and Enrichment This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

	• Draw the line through the points. Problem 2: Graph the quadratic function $y = x^2 - 4x + 3$. • Identify the vertex and axis of symmetry. • Find the y-intercept. • Find the x-intercepts (if any). • Plot the points and draw the parabola. Problem 3 : Problem: Minimizing Cost in Packaging A small business in Davao City is designing a rectangular box with a square base and an open top to package local delicacies. The box must have a volume of 500 cm³. The cost of the material for the base is ₱2 per cm², and the cost for the sides is ₱1 per cm². Task: 1. Let xx be the length of one side of the square base. Express the height in terms of xx. 2. Write a function for the total cost of materials. 3. Find the value of xx that minimizes the cost. 4. Interpret the result.	
Lesson 5. 2		
III. CONTENT	The Function $y = x^3$, $y = \sqrt{x}$, and $y = \sqrt[3]{x}$ (Recommended Time: 2 hours)	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. classify the functions $y = x^3$, $y = \sqrt{x}$, and $y = \sqrt[3]{x}$ based on their equations and graphs as cubic, square root, and cube root functions respectively. 2. sketch the graphs of the given functions with or without the use of technology. 3. apply the functions in solving real-life problems involving growth, motion or geometric dimensions.	

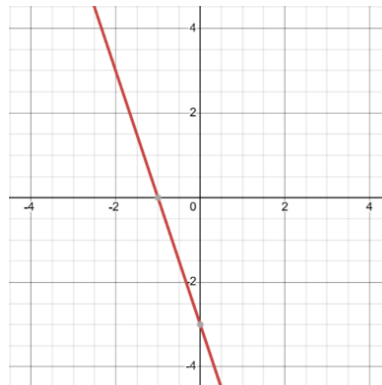
IV. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1: Graph Identification Game Instructions: Identify the following graph of the function, whether it is linear, or quadratic. 1.  Answer: Linear Function 2.  Answer: Quadratic Function	Activity A.1: Graph Identification Game This activity is a quick and interactive way to reinforce learners' ability to recognize and classify various types of function graphs based on their visual characteristics. It supports visual learning and sharpens pattern recognition in mathematical functions, particularly linear, or quadratic. To facilitate this activity, the teacher may choose one from the following suggested strategies. Strategy1: Game-Based Formative Assessment Instructions: 1. Prepare the point-point presentation and screen projector. 2. Divide the class into 4 small groups. 3. Introduce the activity entitled Graph Identification Game 4. Provide the materials needed for the activity such as pen and meta cards. 5. Inform the learner of the mechanics of the game and scoring. 6. Facilitate the discussion by briefly reviewing the guide questions. Strategy2: Think-Pair-Share Instructions: 1. Prepare a set of graph cards or slides showing different types of function graphs 2. Arrange the classroom for easy pairing and sharing. 3. Briefly review key characteristics of different types of functions. 4. Explain the purpose of the activity and the Think-Pair-Share strategy: - Think: Individual reflection - Pair: Discuss with a partner - Share: Present to the whole class 5. Distribute one or two printed graphs to each learner.

3.



Answer: Quadratic Function

4.



Answer: Linear Function

6. Ask learners to individually observe and write down their classification of the function

7. Encourage the learners to justify their answer with at least one characteristic.

8. Have learners' pair-up with a seatmate or assigned partner.

9. Instruct them to:

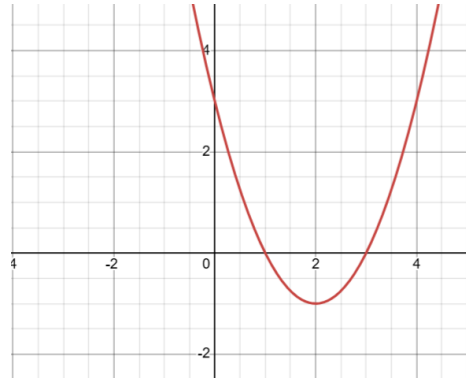
Compare their answers.

Discuss differences or agreements in classification.

Reach a consensus and prepare a short explanation.

10. Invite each pair to share their classification and reasoning with the class.

5.



Answer: Quadratic Function

Guide Questions:

1. What do you notice about the shape of the graph?
2. What makes a function linear?
3. What is the general shape of a quadratic graph?

A.2. Establishing the Purpose of the Lesson

**Activity A.2
Function Match-Up: Real-Life Connections**

Instructions:

There are two sets of cards: one set contains real-life scenarios, and the other set shows different types of functions – linear, and quadratic. Read each scenario carefully and discuss with your group which type of function best fits the situation.

Scenario 1:

Mang Jun sells bangus (milkfish) at ₱180 per kilo. The total cost depends on how many kilos a customer buys.

Answer: Linear Function

Activity A.2

Function Match-Up: Real-Life Connections

To facilitate this activity, the teacher may apply one of the following strategies.

Strategy 1: Match-up Functions

Instructions:

1. Divide the class into four small groups.
2. Provide materials with 5 different scenarios and possible answers.
3. Inform the learners of the mechanics of the activity.
4. Let the learners analyze, discuss and match-up the scenario to the correct functions.
5. Provide an answer sheet.

	Scenario 2: The stream of water from the fountain at Luneta follows a curved path that rises and falls symmetrically. Answer: <u>Quadratic Function</u> Scenario 3: When a player shoots a basketball during the barangay league, the ball follows a parabolic arc. Answer: <u>Quadratic Function</u> Scenario 4: Jane uses ₱3 of load per minute for mobile data. Her remaining balance decreases with time. Answer: <u>Linear Function</u> Scenario 5: A tourist buys strawberry jam bottles at ₱150 each. The total cost depends on how many jars are bought. Answer: <u>Linear Function</u>	Strategy 2: Gallery Walk Instructions: 1. Divide the class into four small groups. 2. Prepare material with 5 different pictures with scenarios. 3. Assign 5 different stations within the classroom only. 4. Provide the mechanics of the activity. 5. The learner will walk in every station, then analyze the scenario given and determine the type of function. 6. Each group has 60 seconds to analyze the problem. Strategy 3: Think-Pair-Share Instruction: 1. Prepare materials for the activity with 5 different scenarios. 2. Arrange the classroom for easy pairing and sharing. 3. Inform the learners of the mechanics of the activity. 4. Distribute one poster with different scenario. 5. Ask learners to individually observe and write down the type of function.
B. Instituting New Knowledge	<i>B.1. Presenting Examples</i> Activity B.1.1: Exploring Functions through Area and Perimeter A farmer wants to fence a square piece of land. He needs to know how much fencing material he will need. If the farmer knows the area of the land, how can he calculate its perimeter? Given a square with side x. Perimeter $P = 4x$ Area $A = x^2$	Activity B.1.1: Exploring Functions through Area and Perimeter This scenario shows that linear and quadratic functions are not always enough to solve the problems we encounter in real life. In this lesson, we will explore other types of functions—such as cubic, square root, and cube root functions—that can help us model more complex situations.

	Guide Questions: 1. Is this function linear? 2. Is it quadratic? 3. What kind of function do you think P(A) is? Activity B.1.2: Function Type Sort Instructions: Sketch the graphs of the following functions using application software like Desmos Graphing Calculator. Then, classify whether the following functions are cubic, square root or cube root. <table><tr><td></td><td style="text-align: center;">Answer</td></tr><tr><td>1. $y = -x^3 + 4$</td><td style="text-align: center;"><u>Cubic</u></td></tr><tr><td>2. $f(x) = \sqrt{x + 3}$</td><td style="text-align: center;"><u>Square Root</u></td></tr><tr><td>3. $g(x) = \sqrt[3]{x}$</td><td style="text-align: center;"><u>Cube root</u></td></tr><tr><td>4. $y = x^3$</td><td style="text-align: center;"><u>Cubic</u></td></tr><tr><td>5. $h(x) = \sqrt[3]{x + 2}$</td><td style="text-align: center;"><u>Cube root</u></td></tr></table> Guide Questions 1. What is the highest exponent of the variable in the function? 2. What are the key characteristics of cubic, square root, and cube root functions? 3. How does the function of a cubic differ from that of a cube root or square root functions? 4. Which function better describes real-world growth and why?		Answer	1. $y = -x^3 + 4$	<u>Cubic</u>	2. $f(x) = \sqrt{x + 3}$	<u>Square Root</u>	3. $g(x) = \sqrt[3]{x}$	<u>Cube root</u>	4. $y = x^3$	<u>Cubic</u>	5. $h(x) = \sqrt[3]{x + 2}$	<u>Cube root</u>	Activity B.1.1: Exploring Functions through Area and Perimeter This activity promotes the use of dynamic graphing tools like Desmos, encouraging technological integration into mathematical visualization. Ensure that learners are guided on how to input each function into Desmos and how to identify visual characteristics of each graph type before classifying. This table serves as an answer guide, but learners should discover the classifications through graph interpretation. To facilitate this activity the teacher may apply ICT integration strategy using Graphing Tool applications. 1. Divide the class into 5 small groups. 2. Prepare the learning materials needed for the activity such as computer laboratory, cellphones, online or offline application software for graphing tool like Desmos. 3. Let the learners explore the graphing of different functions using Desmos and classify the type of function. 4. Facilitate the discussion to answer the guided questions.
	Answer													
1. $y = -x^3 + 4$	<u>Cubic</u>													
2. $f(x) = \sqrt{x + 3}$	<u>Square Root</u>													
3. $g(x) = \sqrt[3]{x}$	<u>Cube root</u>													
4. $y = x^3$	<u>Cubic</u>													
5. $h(x) = \sqrt[3]{x + 2}$	<u>Cube root</u>													
	B.2. Discussing the Concept Activity B.2: Lesson Discussion: The Function $y = x^3$, $y = \sqrt{x}$, and $y = \sqrt[3]{x}$ Instructions: Study the information and examples and prepare for class discussion.	Activity B.2: Lesson Discussion: The Function $y = x^3$, $y = \sqrt{x}$, and $y = \sqrt[3]{x}$ This discussion helps learners to visualize, understand, and analyze the fundamental properties of cubic functions, starting with the simplest form $y = x^3$.												

1. Function: $y = x^3$

The function $y = x^3$ is the simplest form of a cubic function.

Definition:

The function $f: \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$f(x) = x^3,$$

is a cubic function, where each real number input x is mapped to its cube.

Note: by definition of a Function f , we say

$$y = f(x).$$

The cubic function increases at an accelerating rate for positive x , and decreases at an accelerating rate for negative x . It has one inflection point at the origin, where the graph changes concavity.

Key Properties

Domain: All Real Numbers ($\mathbb{R}$) or $(-\infty, \infty)$

Graph:

Sketch the graph of the function $f(x) = x^3$.

Using the table of values, we get

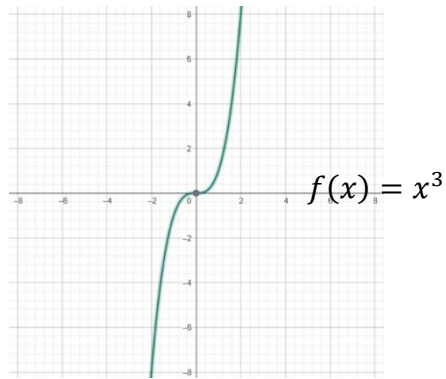
x	-3	-2	-1	0	1	2	3
y	-27	-8	-1	0	1	8	27

By sketching the graph of function $f(x) = x^3$, we have

To facilitate this discussion the teacher may apply one of the following strategies.

Strategy 1: ICT Integration

1. The teacher will create a creative PowerPoint presentation to visualize and catch the learner's attention.
2. Select appropriate tools that align with your lesson goals.



Range: All Real Numbers ($\mathbb{R}$) or $(-\infty, \infty)$

Intercepts:

- x-intercept: (0, 0)
- y-intercept: (0, 0)
- The function passes through the origin

Symmetry:

- Odd function:

$$f(-x) = -f(x)$$
- This means the graph is symmetric about the origin

End Behavior:

- As $x \rightarrow \infty$, then $y \rightarrow \infty$
- As $x \rightarrow -\infty$, then $y \rightarrow -\infty$

Graph Shape:

- An S-shaped curve that passes through the origin, bending upward on the right and downward on the left.

Transformation

General Form of Transformation:

$$f(x) = a(x - h)^2 + k$$

Where:

- a : Affects vertical stretch/compression and reflection over the x-axis

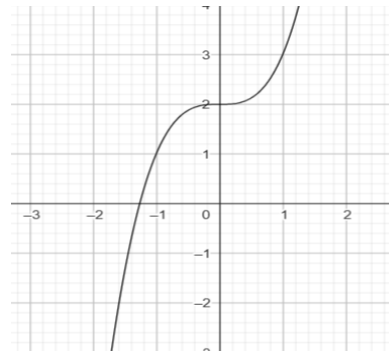
- h : Shifts the graph horizontally (left/right)
- k : Shifts the graph vertically (up/down)

Example of Transformations

1. Vertical Shift

$$f(x) = x^3 + 2$$

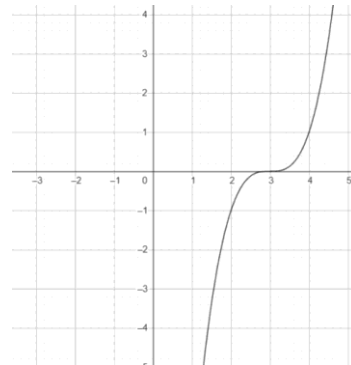
- Moves the graph up 2 units.



2. Horizontal Shift

$$f(x) = (x - 3)^3$$

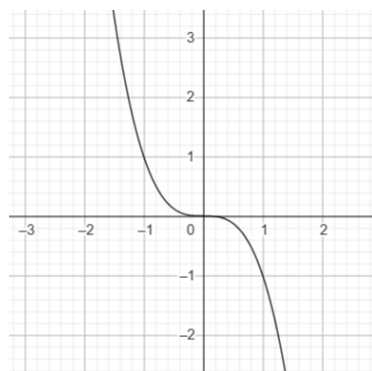
– Moves the graph right 3 units.



3. Reflection Across the x-axis

$$f(x) = -x^3$$

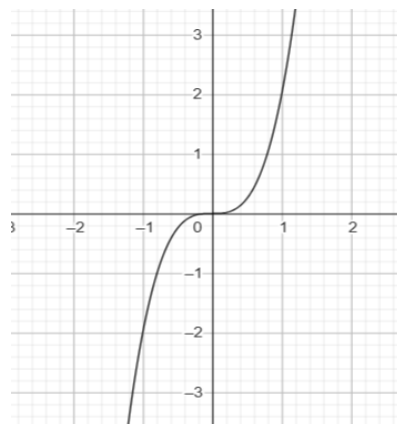
–Flip the graph upside down.



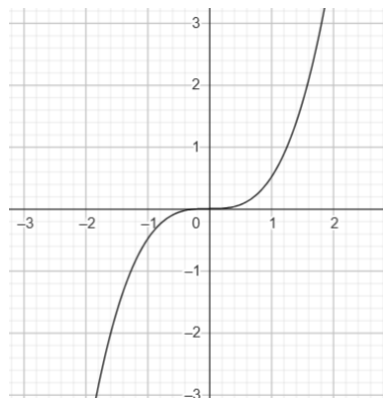
4. Vertical Stretch/Compression

$$f(x) = 2x^3$$

- Stretch the graph taller (steeper)



- Compresses the graph shorter (flatter).



2. Function: $y = \sqrt{x}$

The function $y = \sqrt{x}$ is the simplest form of a square root function.

Definition:

The square root function is defined as:

$$f(x) = \sqrt{x}$$

This means that for each non-negative input x , the function gives the positive number whose square is x .

Key Properties

Domain:

- The domain is $x \geq 0$.
- Since you cannot take the square root of a negative in the real number system, then the function is defined for non-negative real numbers.

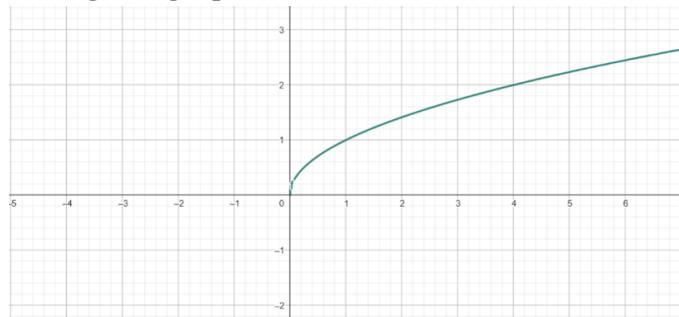
Graph:

Sketch the graph of $y = \sqrt{x}$.

Using the table of values, we get

x	1	2	3	4	5	6	7
y	1.00	1.41	1.73	2.00	2.24	2.45	2.65

By sketching the graph of the function $y = \sqrt{x}$, we have



Range:

- The range is $y \geq 0$.
- The output is always non-negative real numbers.

Intercepts:

- x-intercept: (0, 0)
- y-intercept: (0, 0)
- The function passes through the origin.

End Behavior:

- As $x \rightarrow \infty$, then $y \rightarrow \infty$
- As $x \rightarrow 0^+$, then $y \rightarrow 0$
- The function is increasing on its entire domain $[0, \infty)$. However, it increases slowly – the rate of increase decreases as x increases.

Symmetry

- The function is neither even nor odd.
- It is not symmetric about the y-axis or origin.

Graph Shape:

- Starts at the origin and curves upward.

Transformation

General Form of Transformation:

$$y = a\sqrt{b(x-h)} + k$$

Each parameter transforms the graph in a specific way:

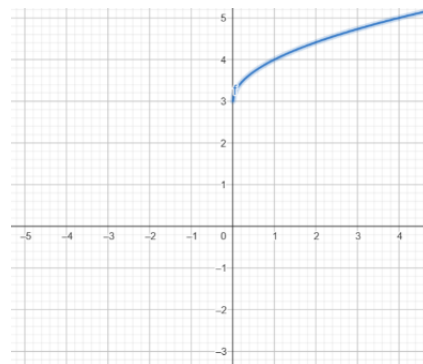
- a : Vertical stretch/compression and reflection over the x-axis
- b : Horizontal stretch/compression and reflection over the y-axis
- h : Horizontal shift (left/right)
- k : Vertical shift (up/down)

Examples of Transformations:

1. Vertical Shift

$$y = \sqrt{x} + 3$$

- Moves the graph up to 3 units.



2. Horizontal Shift

$$y = \sqrt{x - 2}$$

- Moves the graph to the right 2 units.



3. Reflection Across the x-axis

$$y = -\sqrt{x}$$

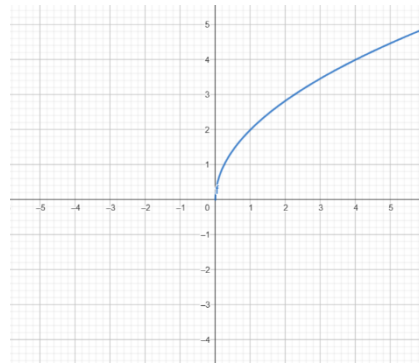
- Flips the graph upside down.



4. Vertical Stretch/Compression

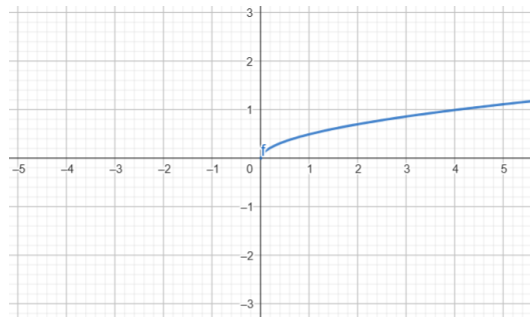
a. $y = 2\sqrt{x}$

-Stretch the graph taller.



b. $y = \frac{1}{2}\sqrt{x}$

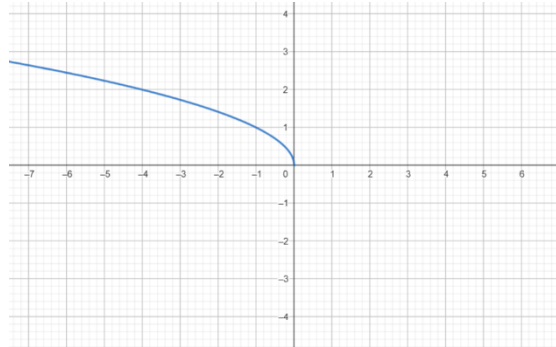
- Compresses the graph shorter.



5. Reflection Across the y-axis (involves inside the root)

$$y = \sqrt{-x}$$

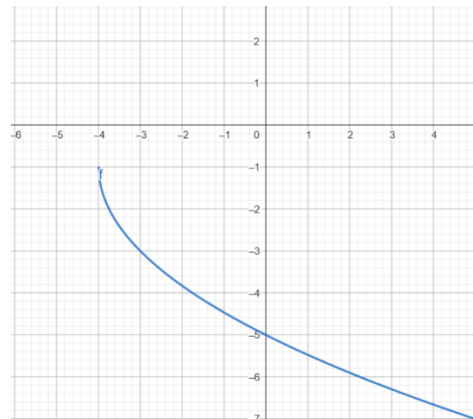
- Reflects over the y-axis. (Note: Domain changes to $x \leq 0$)



6. Combined Transformations

$$y = -2\sqrt{x+4} - 1$$

- Reflect over x-axis, stretch vertically by 2, shift left 4 units, and down 1 unit.



3. Function $y = \sqrt[3]{x}$

The function is the simplest form of the cube root function.

Definition

The cube root function is defined as:

$$f(x) = \sqrt[3]{x}$$

This means that for any real number x , the function gives the number that, when raised to the third power (cubed), equals x .

Key Properties

Domain:

- All real numbers ($\mathbb{R}$) or $(-\infty, \infty)$

Graph:

Sketch the graph of $y = \sqrt[3]{x}$.

Note that $y = \sqrt[3]{x}$ can be express into exponential form $y = x^{\frac{1}{3}}$. By applying cube both sides, we have

$$y^3 = (\sqrt[3]{x})^3$$

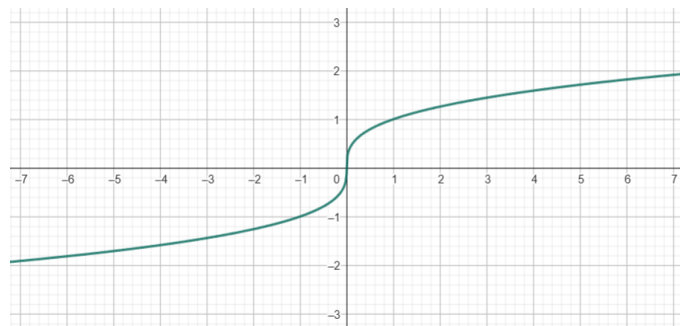
Then

$$y^3 = x.$$

Using the table of values, we get

x	-27	-8	-1	0	1	8	27
y	-3	-2	-1	0	1	2	3

By sketching the graph of the function $y = \sqrt[3]{x}$, we have



Range:

- All real numbers ($\mathbb{R}$) or $(-\infty, \infty)$

Intercept:

- x-intercept: (0, 0)
- y-intercept: (0, 0)
- The function passes through the origin

Symmetry:

- Odd function (symmetric about the origin)

Graph Shape:

- S-shaped curve that increases gradually from left to right

Transformation

General Form of Transformation:

$$y = a\sqrt[3]{b(x-h)} + k$$

Where each parameter affects the graph in the following ways:

a : Vertical stretch/compression and reflection over x-axis

b : Horizontal stretch/compression and reflection over y-axis

h : Horizontal shift: moves the graph left/right

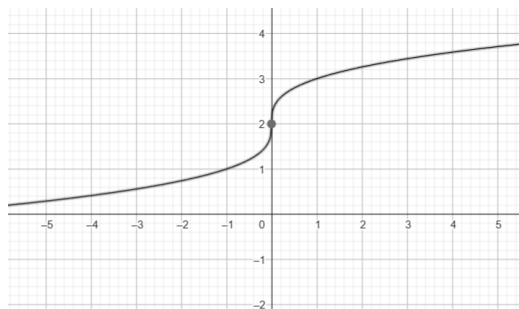
k : Vertical shift: moves the graph up/down

Examples of Transformations:

1. Vertical Shift

$$y = \sqrt[3]{x} + 2$$

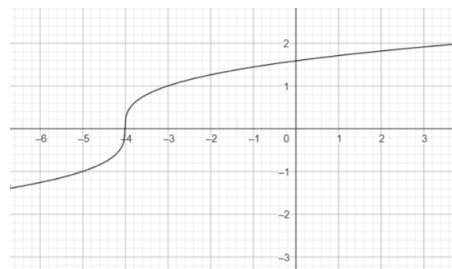
- Moves the graph up 2 units.



2. Horizontal Shift

$$y = \sqrt[3]{x + 4}$$

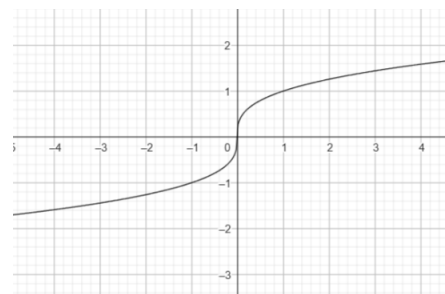
- Moves the graph left 4 units.



3. Reflection Across the x-axis

$$y = \sqrt[3]{x}$$

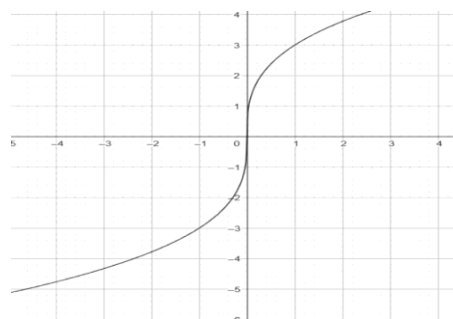
- Flips the graph upside down (reflects over the x-axis).



4. Vertical Stretch/Compression

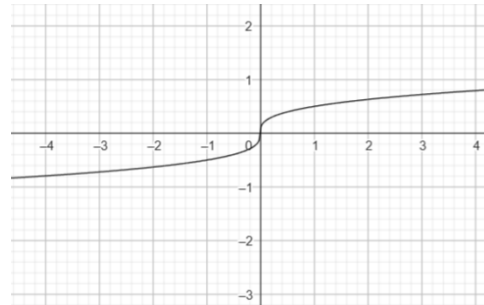
$$y = 3\sqrt[3]{x}$$

- Stretch the graph vertically.



$$y = \frac{1}{2} \sqrt[3]{x}$$

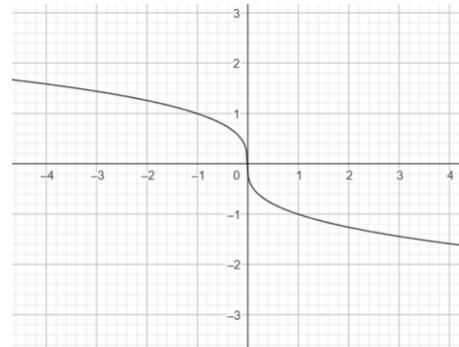
- Compresses the graph vertically (flatter).



5. Reflection Across the y-axis

$$y = \sqrt[3]{-x}$$

- Reflects the graph horizontally over the y-axis.



B.3. Developing Mastery

Activity B.3: Applied Math Practice: Developing Skills on Functions

Instructions:

Sketch the graph using the application software, like GeoGebra Graphing Tool.

1. $y = 5x^3 + 3x^2 + 2x + 5$
2. $y = 2\sqrt{-x}$

Activity B.3:

Applied Math Practice: Developing Skills on Functions

This activity is designed to help learners develop mastery in graph analysis by combining digital tools with manual graphing skills. It bridges technology use with traditional skills and reinforces understanding of different function types through visualization and practice.

To facilitate this discussion the teacher may apply one of the following strategies.

	3. $y = -x^3 + 3x$ 4. $y = \sqrt[3]{x} + 2$ 5. $f(x) = 3\sqrt[3]{x+4} + 2$ Guide Questions • Which graphs have similar shapes or curves? • Do any of the functions have the same domain or range? Explain. • How do the graphs differ in terms of shape? • Do any of the functions show restricted domains? • How does the presence of a square root or cube root affect the direction or openness of the graph?	Strategy 1: Visual-Analytical Pairing Strategy 1. Allow the learners to analyze each function visually using the GeoGebra graph and pair their observations with written explanations. 2. After graphing each function in GeoGebra, identify and write down key features: b. Domain and Range c. Intercepts d. End of behavior e. Symmetry 3. Draw the graph on paper, labeling key points and features. Strategy 2: Think-Pair-Compare Strategy 1. The learners will work individually at first, then collaborate in pairs to compare their hand-drawn graphs and interpretations. 2. Encourage collaborative learning and peer explanation. 3. Each learner will use GeoGebra to graph the functions and completes their manual sketches. 4. In pairs, compare graphs and explanations and discuss differences. 5. Each pair presents one graph to the class, explaining the transformation from the simple form of function.
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Know Your Household Consumptions Problem 1: If the volume V of a cube is related to the side length s by $V=s^3$, how can you determine s given V? Which function would you use? a. Given the Volume V is 25 cm^3, what is the side length s? Problem 2: In a small coastal town in Zamboanga, Aling Nena runs a sari-sari store that sells local brands	Activity C.1: Know Your Household Consumptions This activity aims to help learners understand how mathematical functions—specifically cubic and square root functions—can be applied to real-life situations involving volume and area calculations. It strengthens both algebraic thinking and problem-solving skills by showing how to reverse a function to find an unknown measurement.

	of <i>sardinas</i>. One of the best-selling items is a cylindrical tin of sardines from a brand called “Masarap Sardines”. The label on the can says it contains Volume: 350 cm^3 of sardines. Aling Nena measures the height of the can and finds it to be 10 cm. She wants to stack the cans efficiently on her shelf and needs to know the radius of the base of each can, to calculate how much space she needs. Problem 3: Miguel, a Grade 11 learner from Cebu, wants to surprise his younger sister with a gift. He chooses a small box shaped like a cube, which he plans to wrap with colorful gift wrapper bought from the nearby market. The box has a volume of 64 cm^3, and Miguel wants to know how much wrapping paper he needs to cover the entire surface of the box without leaving any side exposed.	
	C.2. Making Generalization Activity C.2: Graph It Out: Generalizing Function Properties Visually The learners will make a function property diagram based on the assigned function given. They will graph their assigned function using a digital tool like Desmos or GeoGebra. They will write a short summary called a generalization box. In this box, they should list at least three general observations or patterns they noticed about their function. Group 1: $y = x^3 + 1$ Group 2: $y = \sqrt{x}$ Group 3: $y = \sqrt[3]{x}$	Activity C.2: Graph It Out: Generalizing Function Properties Visually This activity guides learners to analyze and summarize function behavior through visual graphing and collaborative reflection. It combines the use of digital graphing tools with critical thinking to help learners construct meaning from patterns they observe in different types of functions. The best approach for creating the summary to be used is the Reflective Learning Strategy.
	C.3. Evaluating Learning	Activity C.3: Checking What We Learned

Activity C.3: Checking What We Learned

Multiple Choice:

Read each question carefully. Choose the letter of the correct answer.

1. Which of the following statements correctly compares the graphs of $y = x^3$ and $y = \sqrt[3]{x}$?
 - A. Both graphs are symmetric with respect to the y-axis.
 - B. Both graphs are increasing for all real values of x.**
 - C. $y = x^3$ decreases while $y = \sqrt[3]{x}$ increases for negative values of x.
 - D. $y = x^3$ has a turning point, but $y = \sqrt[3]{x}$ does not.
2. If a point $(2, a)$ lies on the graph of $y = x^3$, which of the following points will lie on the graph of $y = \sqrt[3]{x}$?
 - A. $(a, 2)$**
 - B. $(-2, -a)$
 - C. $(2, \sqrt[3]{a})$
 - D. $(a, 8)$
3. Which of the following **BEST** explains why $y = x$ is common reference when analyzing the behavior of $y = x^3$ and $y = \sqrt[3]{x}$?
 - A. It is a line of symmetry for both graphs.
 - B. It intersects both graphs only at the origin.
 - C. All three functions pass through the origin and are increasing.**
 - D. It represents the inverse of both $y = \sqrt[3]{x}$ and $y = x^3$.
4. Which of the following real-world situations is best modeled by the function $y = \sqrt[3]{x}$?
 - A. The area of a square as a function of side length
 - B. The volume of a cube as a function of side length

This multiple-choice assessment is designed to evaluate learners' understanding of key properties and relationships among cubic and cube root functions, including their graphs, inverse relationships, and real-world applications. The questions target both conceptual understanding and application skills.

	C. The side length of a cube as a function of volume D. The speed of a car over time in a straight road 5. A learner claims that the inverse of the function $y = x^3$ is $y = \sqrt[3]{x}$. How would you evaluate the correctness of this statement? A. Correct only for positive values of x. B. Incorrect; cubic functions do not have inverses. C. Correct; $\sqrt[3]{x}$ undoes the operation of cubing a number. D. Incorrect; the inverse of a cubic function is a square root function.	
	C.4. Additional Activities Activity C.4: Reinforcement and Enrichment Remediation Instructions: Read and solve each problem. 1. A carpenter in Cebu is making small wooden cubes to teach kids about volume. If each cube has a side length of 4 cm, what is its volume? 2. In Davao, a farmer has a square plot of land with an area of 225 square meters. He wants to know the length of one side. 3. A toy maker in Quezon City has a box with a volume of 1,000 cubic centimeters. What should the length of each side be if the box is a cube? 4. A learner in Iloilo is solving a math problem: Find the value of y if $x=49$ using the square root function.	Activity C.4: Reinforcement and Enrichment This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

	5. A cold storage unit in Baguio is designed in a cube shape. If the side length is 5 meters, what is its volume? Enrichment Read and solve each problem. A group of STEM learners from a senior high school in Butuan is designing a multi-purpose wooden storage box for their community pantry project. The box is cube-shaped and should be able to store rice sacks and canned goods efficiently. The team is given the following specifications: - The volume of the cube must be 216,000 cubic centimeters. - The surface area of one face must be no more than $14,400 \text{ cm}^2$ to fit on a table. - The team also wants to find out how much space per side (in cm) and the diagonal across one face of the box will be. Guide Questions: - What is the side length of the cube box? - What is the area of one face of the box?	
Lesson 5.2		
III. CONTENT	Life in Pieces: Exploring Real-World Scenarios through Piecewise Functions	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: - illustrate absolute value function as a form of piecewise function. - represent a piecewise function in practical contexts (i.e. fare rates, purchasing, and income tax). - solve practical problems involving piecewise functions.	

IV. PROCEDURES	ACTIVITIES	ANNOTATION												
A. Activating Prior Knowledge	A. 1. Activating Prior Knowledge Activity A.1.1: Cracking the Code of Equations of Functions Instructions: Given the following set of equations, determine what type of function each equation represents. <table><thead><tr><th>Equation</th><th>Answer</th></tr></thead><tbody><tr><td>1. $f(x) = 2x + 7$</td><td>Linear</td></tr><tr><td>2. $f(s) = s^2 - 2s - 5$</td><td>Quadratic</td></tr><tr><td>3. $f(t) = \sqrt{x - 1}$</td><td>Square Root</td></tr><tr><td>4. $f(r) = 5r^3 - 9$</td><td>Cubic</td></tr><tr><td>5. $f(u) = \sqrt[3]{2u + 1}$</td><td>Cube Root</td></tr></tbody></table> Activity A.1.2: Match the Scenario Instructions: Given the scenarios, match the type of function that each represents. - Jay wants to know the length of each side of a cube whose volume is 64 cubic cm. Answer: Cube root - The height of an object that Josh has thrown upward. Answer: Quadratic - A passenger who rides a grab and charges a fare with a base fee and cost per kilometer. Answer: Linear - A rectangular box that is formed by cutting equal squares from the four corners of a cardboard and folding up the side.	Equation	Answer	1. $f(x) = 2x + 7$	Linear	2. $f(s) = s^2 - 2s - 5$	Quadratic	3. $f(t) = \sqrt{x - 1}$	Square Root	4. $f(r) = 5r^3 - 9$	Cubic	5. $f(u) = \sqrt[3]{2u + 1}$	Cube Root	Activity A.1: Cracking the Code of Equations of Functions This activity recalls simple properties or characteristics of various functions by means of inspection. This means that learners will just review the fundamental concepts of different functions without going through the tedious solutions. Familiarization with these types of functions helps learners to accurately represent real-world problems. To facilitate the activity, the teacher may choose one from the following suggested approaches. Strategy 1: Think-Pair-Share Instructions for teachers: - The teacher will pair each learner. Note that pairing depends on how the teacher does it. It could be by random selection, seating arrangement, etc. - Each item in the activity shall be flashed on the screen and let all pairs think of the activity. - The teacher will tell the class about the activity called Cracking the Code of Equations and Match the Scenario using the “Think-Pair-Share”. - The teacher shall discuss the purpose of the activity by saying “<i>To classify functions by inspection correctly, you need to understand the key characteristics of functions such as its properties, graphs, behavior. This technique will help you recognize functions easily and accurately.</i>” - To facilitate the activity, the teacher shall:  Tell the learners first to identify the function individually,
	Equation	Answer												
	1. $f(x) = 2x + 7$	Linear												
2. $f(s) = s^2 - 2s - 5$	Quadratic													
3. $f(t) = \sqrt{x - 1}$	Square Root													
4. $f(r) = 5r^3 - 9$	Cubic													
5. $f(u) = \sqrt[3]{2u + 1}$	Cube Root													

Answer: Cubic

5. Finding one side of a square given its area.

Answer: Square Root

✚ Discuss and justify their answers as a pair.

✚ Share responses as a class to clarify misconceptions.

11. The teacher shall summarize all the responses given and provide supplemental discussions when necessary.

Strategy 2: **Function Sorting Station (Rotation Station)**

Instructions:

1. The facilitator shall discuss the purpose of the activity by saying *“To classify functions by inspection correctly, you need to understand the key characteristics of functions such as their properties, graphs, behavior. This technique will help you recognize functions easily and accurately.”*
2. Prepare and print all the questions and have them posted inside the class.
3. Set up stations that contain the questions in activities A and B.
4. The learners move or rotate in pairs in all the designated stations to answer the questions correctly.
5. Instruct learners to work collaboratively and think critically through active manipulations of the questions.
6. Let the learners discuss and share their outputs in the class.
7. The teacher shall process the outputs of the learners.

Strategy 3: **Socratic Method/Argumentative Style**

Instructions:

1. Explain the purpose of the activity by saying *“To classify functions by inspection correctly, you need to understand the key characteristics of functions such as its properties, graphs, behavior. This*

		<i>technique will help you recognize functions easily and accurately.</i> - Write the questions in strips of paper and place them in a box or bowl. - Let the identified pair/s to pick a strip of paper containing the question. - Ask each pair to decode or match the equation/scenario to the corresponding function type. - Require all pairs to DEFEND their answers in WRITING or BRIEF ORAL EXPLANATIONS.																		
A.2. Establishing the Purpose of the Lesson Activity A.2: Check Me Out: Function or Not Instruction: There are provided conditions in the table and identify whether each item represents a function or not. <table border="1" data-bbox="459 826 1211 1520"> <thead> <tr> <th>Examples</th><th>Function or not?</th><th>Reason</th></tr> </thead> <tbody> <tr> <td>1. $f(x) = x + 2$</td><td>Function</td><td>One input yields one output</td></tr> <tr> <td>2. $x^2 + y^2 = 4$</td><td>Not</td><td>Vertical Line hits the graph twice</td></tr> <tr> <td>3. $y = x^2$</td><td>Function</td><td>Vertical Line hits the graph once</td></tr> <tr> <td>4. $1 \rightarrow A, 1 \rightarrow B, 1 \rightarrow C$</td><td>Not</td><td>One-to-many</td></tr> <tr> <td>5. $f(x) = \begin{cases} x + 3, & x < 0 \\ 3x, & x \geq 0 \end{cases}$</td><td>Function</td><td>Still one output per input</td></tr> </tbody> </table>		Examples	Function or not?	Reason	1. $f(x) = x + 2$	Function	One input yields one output	2. $x^2 + y^2 = 4$	Not	Vertical Line hits the graph twice	3. $y = x^2$	Function	Vertical Line hits the graph once	4. $1 \rightarrow A, 1 \rightarrow B, 1 \rightarrow C$	Not	One-to-many	5. $f(x) = \begin{cases} x + 3, & x < 0 \\ 3x, & x \geq 0 \end{cases}$	Function	Still one output per input	Activity A.2: Check Me Out: Function or Not To facilitate this activity the teacher may apply one of the following strategies. Strategy 1: Comparison and Contrast Instructions for teacher: - Divide the class in five groups and ask them to identify the given conditions as to function or not. - Let them recall some properties of functions or their graphs or the Vertical Line Test. - Instruct the groups to provide reason/s of their responses (Responses may vary). - Explain to the class that this activity leads to the basic property of piecewise as a function even if there is discontinuity/break/sharp corner/cusp in its graph. Strategy 2: Stand and Vote Instructions for Teachers - Print or post each item containing the condition. Each group draws one item at a time. - Begin the activity by saying: “Let’s do a quick game to see what we already know about each type of functions and its properties”
Examples	Function or not?	Reason																		
1. $f(x) = x + 2$	Function	One input yields one output																		
2. $x^2 + y^2 = 4$	Not	Vertical Line hits the graph twice																		
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4. $1 \rightarrow A, 1 \rightarrow B, 1 \rightarrow C$	Not	One-to-many																		
5. $f(x) = \begin{cases} x + 3, & x < 0 \\ 3x, & x \geq 0 \end{cases}$	Function	Still one output per input																		

		- Briefly ask for 1–2 volunteers to explain or reason out why the item is a function or not. - Connect to the lesson by stating: “This activity will help you visualize those functions are not only continuous, they are also discontinuous or with a break.
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: Ride the Fare Curve Instructions: Read and answer the questions: Prompt Questions: - Have you ever used a Grab or Angkas while in Manila? - Have you wondered how the fares are calculated? - How much will you pay given the trips below. Suppose you ride in a Grab with service charges of P50.00 flat rate for the first 3 km, an additional P15.00 per kilometer for the next 10 km (from 3rd km to 13th km), and P20.00 per kilometer for any distance travelled beyond 13 km. How much will you pay for a 2-km, 2.5-km, 4.8-km, 10-km, 12.7-km, 15-km, 18.2-km trips?	Activity B.1: Ride the Fare Curve The following are the suggested strategies to present or deliver the activity. Strategy 1: Investigative Approach Instructions: - Divide the class into small groups and let them visualize or reflect on the prompt question and the scenario question shown on the screen. - Give time for the groups to write down essential details of the scenario. - Tell the groups to: ✚ Break down the fare structure/pricing rule into segments/pieces. ✚ Calculate the fare cost for each given distance (manually or technology-based). The trip distance may include fraction/decimal (i.e. 2.5 km, 4.8 km, 12.7 km, 18.2 km, etc.) ✚ Sketch these values obtained with x – axis as the trip distance and y – axis as the distance travelled. ✚ Ask: How much will be paid for the first 3 kms? From 3rd to 13th kms? And beyond 13 kms?

✚ From these responses from the learners, the teacher will present each sub-function into one function which is in piecewise form.

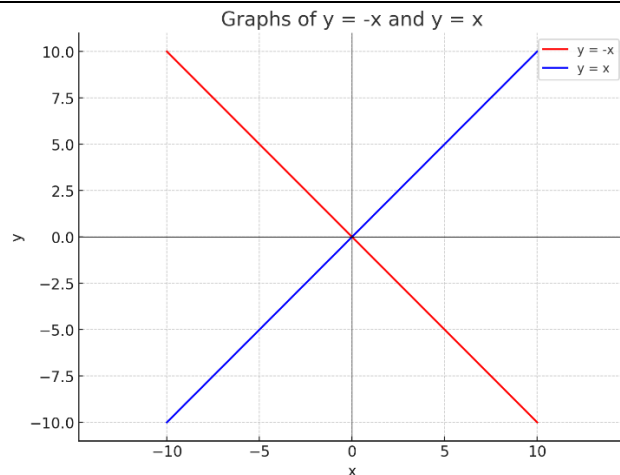
4. Connect the activity to the lesson by saying *“Fare structures/pricing principles can be understood or modeled using functions. This lesson helps you explore and learn more about another type of function which is a **Piecewise Function.**”* This function is usually applied in shipping charges, bus/taxi fare, mobile data plan, parking lot rates, etc.

Strategy 2: Game-based Instruction

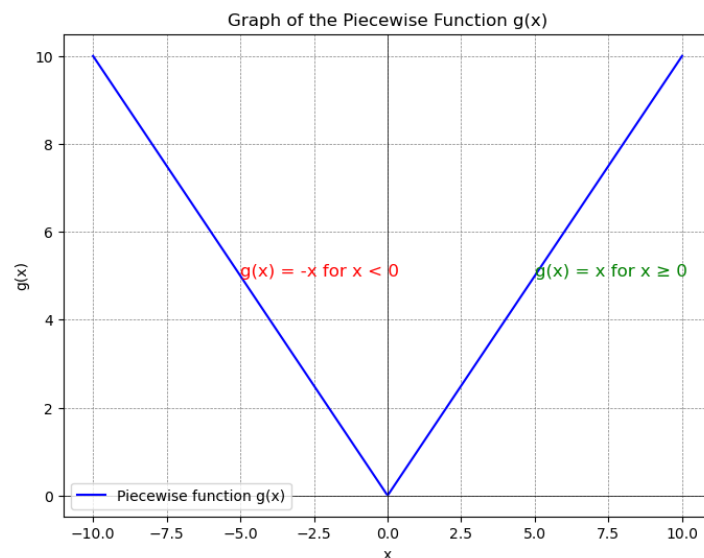
Instructions:

1. Set up the following materials
 - ✚ Printed trip distance cards containing the following distances – 2 km, 8.5 km, 10.3 km, 15.4 km, 18.2 km, etc.
 - ✚ Provide an individual score sheet.
 - ✚ Designate a time keeper.
2. Tell the goal of the activity, that is, to be the first group-player to correctly answer the Grab’s fare for each trip card.
3. In a slate board, each team’s final answer will be raised and checked by the teacher.
4. Each group will be asked to explain or justify their answers why it was so. The group with the correct rationalization or justification of the solutions shall be merited with the highest possible score.
5. Use the scoring guide to appraise the group’s output.
 - Fastest correct answer – 5 points
 - Correct but late – 3 points
 - Incorrect – 0 point

		6. Each question shall be allotted for 1 minute to answer. Connect the activity to the lesson by saying “<i>Fare structures/pricing principles can be understood or modeled using functions. This lesson helps you explore and learn more about another type of function which is a Piecewise Function.</i>” This function is usually applied in shipping charges, bus/taxi fare, mobile data plan, parking lot rates, etc.
	B. 2. Discussing New Concept Activity B.2: Lesson Discussion: Understanding Piecewise Function Instructions: Listen and participate to the dicussions. Key Terms and Definitions: <i>Absolute Value Function.</i> A function that shows the distance of a number from zero on the number line and denoted by the form $f(x) = x$. Consider the function defined by $g(x) = x$ In fact, the function above is analogous to a piecewise function written as: $g(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$ To recognize how the graph looks like, we recall and use the linear function. If $g(x) = -x$ and $g(x) = x$, we obtain the graph	Activity B.2: Lesson Discussion: Understanding Piecewise Function To facilitate the discussion, teacher may select any of the following strategies: Strategy 1: Technology-Assisted Discussion Instructions: 1. Display on the screen each individual task to be assigned to every group. 2. Maximize the use of technology such as graphing calculators, Geogebra, etc. 3. Each group will plug-in the functions. 4. Gather their outputs and post them. 5. Let each representative to explain briefly about the output. Strategy 2. Graph-Equation Matching Instructions for Teachers: 1. Use a small group to facilitate the activity. 2. Give learners/groups graphs of piecewise functions and have them match or write corresponding equations—and vice versa.



Since, $g(x) = -x$ and $g(x) = x$ is expressible as $g(x) = |x|$, then, we only consider the positive values of $g(x)$. Hence, its graph is



Another function is defined by

$$f(x) = |x - 1|$$

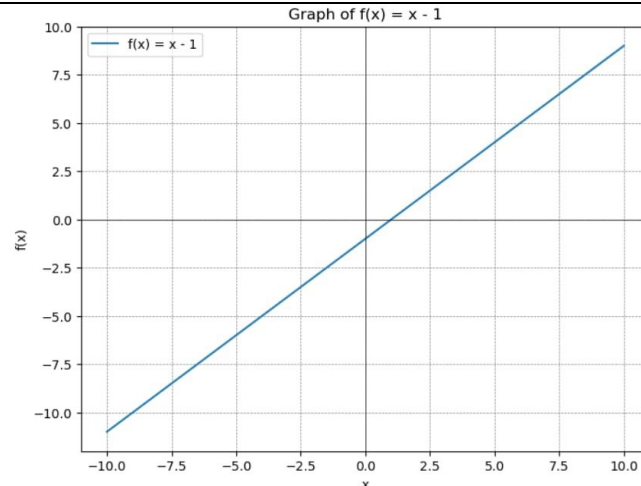
To graph, we know that $f(x) = x - 1$ is a linear function whose graph is

3. Require them to recall some basic properties of functions.
4. Ask them to provide justification about the matching response they made.
5. Conduct FGDs among the groups to summarize the discussions.

Strategy 3. **Game-Based Activities**

Instructions for Teachers:

1. Utilize the same group, provide the goal/s of the activity.
2. Use interactive games (e.g., "Fare Wars" or card sorting) where learners compute fares, match graphs, or solve challenges involving breakpoints.
3. Provide a scoring rubric to evaluate the output of the groups.
4. Conduct round table discussions to clear up things that went right and wrong.
5. Provide supplemental discussions when necessary.

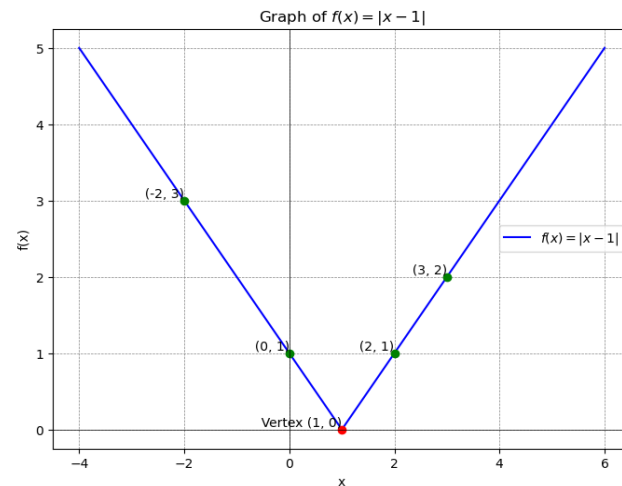


Now, to visualize the graph of $f(x) = |x - 1|$, we keep the positive y values and flip upward the negative values which eventually looks a V-shape. Thus, $f(x) = |x - 1|$ has the graph of

Thus, the function $f(x) = |x - 1|$ can be rewritten as $f(x) = -(x - 1)$ for $x < 1$ and $f(x) = (x - 1)$ for $x \geq 1$. In other words,

$$f(x) = |x - 1|$$

is



equivalent to a piecewise function denoted by:

$$f(x) = \begin{cases} -(x-1), & x < 1 \\ (x-1), & x \geq 1 \end{cases}$$

Remarks:

Recall also that replacing x by $x - 1$ translates the graph of $f(x) = |x|$ one unit to the right. Therefore, an absolute value function can be understood as a form of ***piecewise function***.

How about these? Graph each and write its corresponding piecewise function.

1. $y = |5x + 2|$
2. $y = |x^3|$
3. $y = |x^2 - 1|$
4. $y = |x| + 1$
5. $y = 2|x + 3|$

So, what is Piecewise Function?

Piecewise Function is a function that uses multiple functions, each applying to a specific interval of x .

Here's an example of piecewise function.

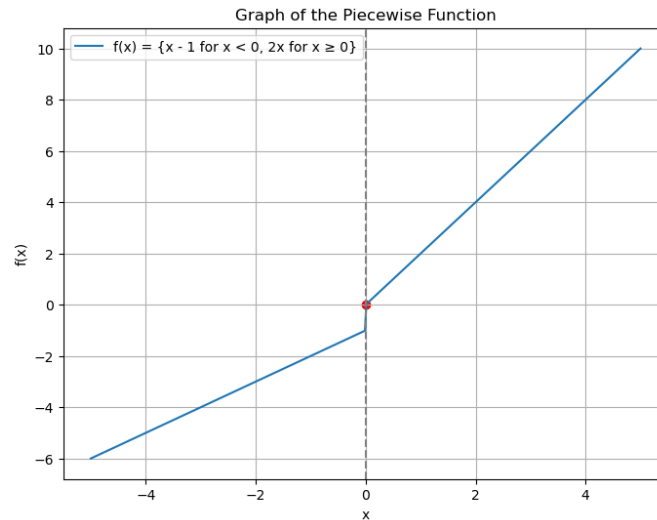
Example 1.

$$f(x) = \begin{cases} x - 1, & x < 0 \\ x, & x \geq 0 \end{cases}$$

This indicates that

- For all values of x less than 0, we apply $x - 1$.
- For all values of x greater than 0, we apply x .

Since the sub-functions are both linear, we expect a combination of two lines. The graph is shown below.



Example 2.

James is enrolled in driving school that requires a rental car fee of P 200. 00 for the first 5 km. and for every kilometer added charges an additional fee of P 40.00. Express a piecewise function for the problem.

Answer: $f(x) = \begin{cases} 200, & 0 < x \leq 5 \\ 200 + 40(x - 5), & x > 5 \end{cases}$

Example 3.

Alfred, a lemon farmer, sells ripe lemons either per piece or bulk. He sells lemons at P15.00 each for orders less than 40 pieces and P550.00 for a bulk of 40 pieces and P10.00 for each excess lemon after that. Write the required piecewise function.

Answer: $f(x) = \begin{cases} 15x, & 0 < x < 40 \\ 550, & x = 40 \\ 550 + 10(x - 40), & x > 40 \end{cases}$

Example 4.

Internet Provider A charges your phone plan differently depending on how much data you consume. The first 3GB is for free, the next 4 GB charges P50.00/GB, and anything consumed beyond that is P70.00/GB. How would you represent this mathematically?

Answer: $f(x) = \begin{cases} 0, & 0 \leq x \leq 3 \\ 50(x - 3), & 3 < x \leq 7 \\ 200 + 70(x - 7), & x > 7 \end{cases}$

Example 5.

Mobile phone plan costs P1, 000.00 per month. The plan composes of 100 free minutes and charges P 7.00 for each succeeding minutes of usage. Solve the monthly bill incurred if the owner used the following minutes. Use table of values and construct the graph.

1. 80 minutes
2. 130 minutes
3. 150 minutes

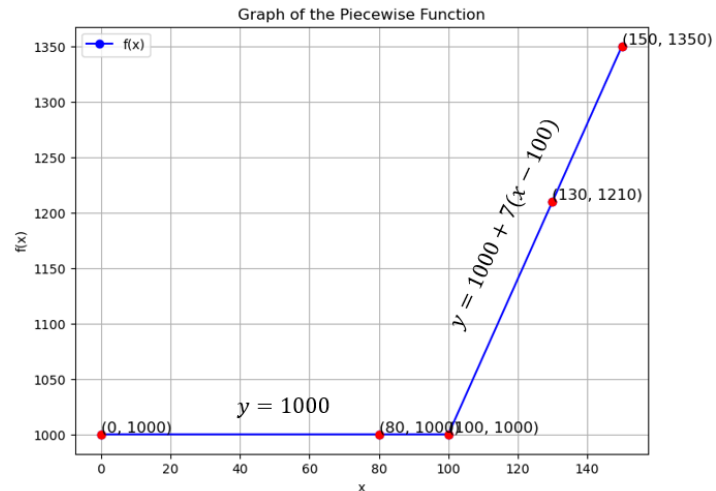
Answer.

Let f be the cost monthly mobile plan and x be the number of minutes used. This can be expressed as piecewise function

$$f(x) = \begin{cases} 1000, & 0 \leq x \leq 100 \\ 1000 + 7(x - 100), & x > 100 \end{cases}$$

x	$y = f(x)$	<i>Point</i>
0	1000	(0,1000)
80	1000	(80,1000)
100	1000	(100,1000)
130	1210	(130,1210)
150	1350	(150,1350)

Graph:



B.3. Developing Mastery

Activity B.3: Delving Deeper

Exercise 1. Sketch the Graph of the indicated functions.

1. $t(u) = \begin{cases} 100, & 0 < u \leq 8 \\ 100 + u, & u > 8 \end{cases}$
2. $P(x) = \begin{cases} x + 1, & \text{if } x < 3 \\ (x - 4)^2 + 1, & \text{if } x \geq 3 \end{cases}$
3. $f(x) = \begin{cases} x - 1, & \text{if } x \leq 2 \\ x^2, & \text{if } x \geq 2 \end{cases}$
4. $h(s) = \begin{cases} 1, & s < -4 \\ s - 1, & -4 \leq s \leq 4 \\ -2s + 4, & s > 4 \end{cases}$

Exercise 2. Express the problems into piecewise function and solve.

1. ABC Hotel has a special promo for their executive room (maximum of two people). The owner charges

Activity B.3: Delving Deeper

Teacher may apply any of the following strategies.

Strategy 1. Presenter-Reactor Feud

Instructions:

1. Use a small group to facilitate the activity.
2. Provide each group a copy of the task. From each group, there shall be a presenter and a reactor.
3. Assign a presenter who shall present the output of the group and shall discuss and explain clearly the final output.
4. Instruct the remaining groups or non-presenting groups to designate reactors who shall task to provide feedback with regard to the presenter's discussion and point out areas of improvement.
5. Assign a moderator to facilitate the presenter-reactor discussion.
6. Conduct FGDs among the groups to summarize the discussions.

Strategy 2. Think-Pair-Solve-Share

Instructions:

1. Give each learner a set of problems.

	P1, 800 for at most three nights stay and P 400 for each additional night of stay. Suppose P is the total cost in terms of the number of nights r that visitor stays. a. Define the piecewise function; b. Find $P(3), P(6), P(10)$. c. What do your answers in part (b) show? 2. Given that a community tax states as indicated below: <table><tr><th>Gross Income Annually (x) in Pesos</th><th>Community Tax Due</th></tr><tr><td>$x \leq 100,000$</td><td>P 0.00</td></tr><tr><td>$100,000 < x \leq 200,000$</td><td>2% of income over P 100,000</td></tr><tr><td>$x > 200,000$</td><td>7% of income over P 200,000</td></tr></table> a. Model the given data using a piecewise definition function. b. Compute the community tax due from the annual gross earnings of c.1. P 50, 590. c.2. P 178, 900. c.3. P 340, 500. c. Illustrate the data in part (c) using graph.	Gross Income Annually (x) in Pesos	Community Tax Due	$x \leq 100,000$	P 0.00	$100,000 < x \leq 200,000$	2% of income over P 100,000	$x > 200,000$	7% of income over P 200,000	2. Ask them to solve one problem alone (Think). 3. Then, pair them with a seatmate to discuss and solve the next one together (Pair + Solve). 4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share). 5. You may use visual aids like posters or digital slides to show each problem. Strategy 3. Interactive Whiteboard Relay Instructions: 1. Divide the class into two teams. 2. On the board/projector, display one problem at a time. 3. A learner from each team races to solve and explain their answer on the board. 4. Award points for correct answers and clear explanation. 5. Encourage class discussion after each round—ask if anyone solved it differently or spotted errors
Gross Income Annually (x) in Pesos	Community Tax Due									
$x \leq 100,000$	P 0.00									
$100,000 < x \leq 200,000$	2% of income over P 100,000									
$x > 200,000$	7% of income over P 200,000									
C. Demonstrating Knowledge and Skills	C.1.Finding Practical Application Activity C.1: Piecewise in the Wild To make piecewise functions real-life, capture some activities with a relatable situation that involves different rules depending on the condition—like fare rates, mobile data charges, or electricity billing. These real-life contexts make the abstract concept meaningful and relevant to learners. Activity: Let’s get it on!	Activity C.1: Piecewise in the Wild Instructions: 1. Divide the class into small groups, say group of four. 2. Each group shall be provided with real-world scenario to work with the team. 3. The four key themes of the activity shall include: 🚖 Taxi fare								

	Instructions: Create a piecewise function for the following. Scenario 1. Taxi Fare • Base fare: ₱35 for the first 1.5 km • ₱8 per km for the next 10.5 km • ₱20 per km beyond 15.5 km Scenario 2. Overtime pays • Regular pay: ₱80/hour for up to 6 hours • Overtime: ₱100/hour beyond 6 hours Scenario 3. Parking Fees • P100 for the first hour • P75 for each additional hour up to 5 hours • Flat P500 for more than 5 hours Scenario 4. Water Bills • First 10 m³: ₱20/m³ • Next 20 m³: ₱30/m³ • Beyond 30 m³: ₱50/m³	 Overtime pays  Parking Fees  Water billing 4. Ask the learners to describe the scenario and discuss among their teammates. 5. The learners shall define the piecewise function 6. They will graph their output and explain their work briefly.
	C.2. Making Generalization Activity C.2: Learning Ticket: My Trip Through the Lesson Instructions: Imagine the lesson as a journey. Each learner fills out their "ticket" as a reflection tool.	Activity C.2: Learning Ticket: My Trip Through the Lesson To wrap-up the lesson, ask each learner fills out their "ticket" as a reflection tool.

	1 •Best Stop • The part of the lesson that made the most sense was... 2 •Bumpy Road • The part of the lesson i have trouble with was... 3 •Funny Moment •The part of the activity that i enjoyed the most was.. •One Thing I'll Remember •If I would explain this topic to a friend, I would say that...	
	C.3. Evaluating Learning Activity C.3: Checking for Mastery Instructions: Read and solve the following problems. 1. A certain Liquor is sold for P150.00 each. With an increasing public demand, a vendor decides to sell it for P140.00 each if someone buys more than 15 bottles. a. Express the cost T with respect to the number x of bottled liquors sold. b. Calculate the $T(8), T(15), T(20)$. c. Draw the graph. 2. Rental cars charge P100.00 for the first three hours and an excess of P20.00 for each hour (or a fraction of it) after that. If you rent a car for more than ten hours, a fee of P500.00 shall be charged. a. Represent the rental car fee in piecewise function b. How much will be paid if the rental time are as follows (a). 1.8 hrs., (b) 4.2 hrs., (c) 12.7 hrs. c. Explain briefly your answer.	Activity C.3: Checking for Mastery Each learner should complete this activity on their own. It plays a key role in evaluating how well they understand the lesson. Be sure to check their work right away so you can offer quick and suitable help to those who may need extra support or reinforcement.

3. A tricycle ride costs ₱10.00 for the first 3 kilometers, and each additional full kilometer adds ₱2.00 to the fare.
 - a. Let d be the distance (in kilometers) of the ride. Use a piecewise function to represent the total fare $f(d)$ in terms of d .
 - b. Find the total fare $f(d)$ if $d = 1.8, 2.1, 3.4, 4.8$.

C.4. Additional Activities

Activity C.4: Reinforcement and Enrichment

Remediation: Scaffolding Learning

Rewrite the given scenario in a piecewise function.

1. Data usage charges can be tiered:
 - Free up to 2 GB
 - 150 per GB for 2–10 GB
 - 300 per GB beyond 10 GB
2. Utility companies charge based on usage tiers:
 - P 5.00 per kWh for the first 80 kWh
 - P 7.50 per kWh for the next 160 kWh
 - P 10.00 per kWh for usage above 240 kWh
3. Shipping fees often depend on weight:
 - P 50.00 for packages up to 1 kg
 - P 100.00 for 1–5 kg
 - P 200.00 for 5–10 kg
 - P 500.00 for over 10 kg

Enrichment: Deepening the Core of Understanding

Solve the given problems. Show your complete solutions.

Situation 1: A company pays its sales representatives based on a piecewise function. The salary structure is defined as follows:

- ✚ If a salesperson's total sales x is less than P100,000, they receive a base salary of P5,000 (no commission).

Activity C.4: Reinforcement and Enrichment

This extra activity can be assigned as homework. Provide remediation for learners who haven't fully mastered the lesson, and offer enrichment tasks to those who are ready to advance their learning further.

- ✚ If their total sales x is P100,000 or more, they receive a base salary of P5,000 plus a 10% commission on all sales.

Define the salesperson's total earnings, $g(x)$, as a piecewise function based on their total sales x

1. Write the piecewise function for $g(x)$.
2. Calculate the total earnings for a salesperson who made P80,000 in sales.
3. Calculate the total earnings for a salesperson who made P150,000 in sales.

Situation 2: A company offers two different pay structures for its sales representatives:

1. **Plan A:** A fixed base salary of **P5,000** with **no commission** for sales below **P10,000**. If sales reach **P10,000 or more**, the salesperson earns **P5,000 plus a 10% commission on all sales**.
2. **Plan B:** A **lower base salary of P3,000** but a **15% commission on all sales**, regardless of the amount.

Define the total earnings function $f(x)$ for **Plan A** and $g(x)$ for **Plan B**.

Questions to Solve:

1. Write the piecewise function for $f(x)$ (Plan A) and the linear function for $g(x)$ (Plan B).
2. Determine which plan provides better pay for **sales of P8,000**.
3. Determine which plan provides better pay for **sales of P15,000**.
4. When is Plan A better than Plan B?

V. ASSESSMENT	For both written and performance tests, the unit summative assessment is provided as an attachment to this lesson exemplar.
VI. REFLECTION	Teacher Reflection Journal. A Navigation Around Piecewise Function As you travel back to your lecture, take time to figure out not only the things that happened but also the instances where you built happy moments with your learners. You can state the following. • Moments that Mattered • Roadblocks and Detours • Navigating the Challenges • Looking Ahead.

Summative Assessment

Instructions:

Answer the unit assessment that includes items from the previous lessons on linear, quadratic, square root, and cube root functions

Exercise 1. Multiple Choice. Read each question carefully. Choose the letter of the correct answer.

1. What is the shape of the graph of a quadratic function?
 - A. Line
 - B. Circle
 - C. Parabola
 - D. Triangle

2. What is the y-intercept of the line $y = -2x + 5$?
 - A. -2
 - B. 5
 - C. -5
 - D. 2

3. Which of the following statements correctly compares the graphs of $y=x^3$ and $y=\sqrt[3]{x}$?
 - A. Both graphs are symmetric with respect to the y-axis.
 - B. Both graphs are increasing for all real values of x.
 - C. $y=x^3$ decreases while $y=\sqrt[3]{x}$ increases for negative values of x.
 - D. $y=x^3$ has a turning point, but $y=\sqrt[3]{x}$ does not.

4. If a point $(2,a)$ lies on the graph of $y=x^3$, which of the following points will lie on the graph of $y=\sqrt[3]{x}$?
 - A. $(a,2)$
 - B. $(-2,-a)$
 - C. $(2,\sqrt[3]{a})$
 - D. $(a,8)$

5. Which of the following BEST explains why $y=x$ is common reference when analyzing the behavior of $y=x^3$ and $y=\sqrt[3]{x}$?
 - A. It is a line of symmetry for both graphs.
 - B. It intersects both graphs only at the origin.
 - C. All three functions pass through the origin and are increasing.
 - D. It represents the inverse of both $y=\sqrt[3]{x}$ and $y=x^3$.

Exercise 2. Solve the given problems.

1. Sketch the graph of $f(x)=5x-2$ on the given domain and determine the range.

$$\{x \mid x=-3,-2,-1,0,3,5,7\}$$

$$\{x \mid -3 \leq x \leq 8\}$$

$$\{x \mid x < -1 \cup x \geq 5\}$$

$$\{x \mid x \in \mathbb{R}\}$$

2. Sketch the graph of $t(u)=-u^2+7u-10$ on the interval $[-2,8]$.

3. Sketch the graph of $r(x)=-\sqrt{(x+4)}$.

4. Sketch the graph of $f(x)=\begin{cases} x+2, & 0 < x < 3 \\ 2, & x \leq 0 \\ x^2-4, & x \geq 3 \end{cases}$

Exercise 3. Solve the following problems. Show your complete solutions.

1. A taxi fare initially charges P 30.00 for the first 4 km distance travelled and P 8.50 for each succeeding kilometer over the initial fare.
 - a. Determine the fare function for $x > 4$.

b. What is the fare cost for a distance of 50 km?

2. Teacher Bryan runs a milk tea café which follows an operating cost represented by the objective function $N(k) = (k-3)^2 + 10$, where $N(k)$ is the incurred daily cost to produce k cup of milk tea.

a. Graph $N(k)$.

b. How many cups of milk tea should Teacher Bryan sell to produce a minimum cost?

c. What is the minimum cost?

3. Electricity providers in the Philippines often use a tiered billing system. The more electricity you consume, the higher the rate per kilowatt-hour (kWh). A billing structure is as follows:

For the first 50 kWh, the rate is ₱6 per kWh.

For the next 100 kWh (from 51 to 150 kWh), the rate is ₱8 per kWh.

For any usage above 150 kWh, the rate is ₱10 per kWh.

a. Define a piecewise function that models the total electricity bill.

b. Create a table showing the total bill for selected values such as 30, 75, 150, 200.

c. Graph the function to visualize how the bill increases with usage.

Performance Task: Investigating Electricity Billing

Scenario:

Electricity bills in the Philippines are often based on a tiered pricing system, where the cost per kilowatt-hour (kWh) increases with higher usage. However, the exact structure is not always clear to consumers.

Your Task:

You are an energy analyst tasked with understanding how electricity billing works in your area. Your goal is to model the billing system using a piecewise function.

Steps:

1. Research or collect data from actual electricity bills (your own, family members, or sample bills online). Identify how the total bill changes with electricity usage (in kWh).
2. Analyze the data to determine if there are different pricing tiers (e.g., a base rate, then higher rates for higher usage).
3. Define a piecewise function ($E(x)$) that models the total cost of electricity based on usage (x) in kWh.
4. Graph the function to visualize how the cost increases with usage.

Reflect and discuss:

1. What strategies could help reduce electricity costs?
2. Present your findings in a short report or presentation.

Rubric: This can be modified.

Criteria	Excellent (4 pts.)	Good (3 pts.)	Satisfactory (2 pts.)	Needs Improvement (1 pt.)
Data Collection and Analysis 20%	Collected accurate and relevant data; clearly identified pricing tiers	Collected data and identified most pricing tiers	Collected some data; pricing tiers unclear or incomplete	Little or no data collected; no clear analysis
Piecewise Function 30%	Function is accurate, complete, and clearly written	Function is mostly accurate with minor errors	Function has some errors or missing parts	Function is incorrect or missing
Graph 40%	Graph is accurate, labeled, and clearly shows piecewise behavior	Graph is mostly accurate with minor issues	Graph is present but has major errors or is unclear	Graph is missing or incorrect
Reflection and Discussion 5%	Provides an insightful analysis of fairness and impact, with well-supported points.	Provides a good analysis of fairness and impact, with some supporting points.	Provides a basic reflection on fairness and impact, with limited support.	Provides little or no reflection or analysis of fairness and impact.
Presentation 5%	Clear, organized, and visually appealing	Mostly clear and organized	Somewhat disorganized or unclear	Disorganized or incomplete